

COUNTING THE ACTIVE DIMENSION OF A TRAINING RUN: WHAT A SINGLE SCALAR LOG CAN AND CANNOT MEASURE

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ABSTRACT

Over a window of training, a neural network’s parameter trajectory moves on a set far smaller than the parameter space. We call the dimension of that set the active dimension of the window: a number of independent components, distinct from the number of directions the optimiser may move in, from the rank of the map to the outputs, and from the effective rank of the trajectory covariance. A stored trajectory measures it directly but is expensive, where a scalar log costs nothing, so we ask what survives the projection onto one. The instrument is standard, delay reconstruction followed by a maximum likelihood intrinsic dimension, and its hazard is that it returns a plausible number when no invariant set exists. On six constructed systems, ending with a linear head on a frozen image classifier, it recovers that number on withheld ranks and seeds to within about one component, up to a ceiling near eight directions; two diagnostics computed from the series alone identify the regimes in which its value may not be read as the active dimension. Applied to grokking they place both standard settings outside the admissible regime. Paying instead for the trajectory, the effective rank collapses near generalisation in four regularised runs and then re-expands, the simplification the mechanistic accounts of grokking lead one to expect; their controls decline too, so what marks the transition is not its depth but that it is sharp, locked to the transition and reversed. Re-run at a window matched to that transition rather than to the record, the log estimate drops at the same step in the same four runs and in neither control, by more than surrogates preserving the observer’s shape produce: two signatures of one event, the second already written by the run.

1 INTRODUCTION

A neural network is trained in a parameter space of dimension P , but the trajectory the optimiser follows does not fill it. Over a window it moves on a much smaller set, and that set grows and shrinks over a run. We call its dimension the *active dimension* of the window (equation (1)). It is a count of independent components: if the optimiser is carried by r independent quasiperiodic phases the orbit closes onto an r -torus and the active dimension is r , with no tunable scale. It is not the dimension of the subspace the optimiser is confined to, which the parameterisation fixes (Li et al., 2018). It is not the rank of the map from that subspace to the outputs. And it is not the effective rank of the trajectory covariance, which anisotropy depresses below it. Section 3.1 separates the four.

Knowing it would settle questions currently argued with whichever of those four quantities is least expensive to compute: whether an apparent simplification of training is a real reduction in degrees of freedom or an artefact of a shrinking scale, and whether a phase transition can be seen without instrumenting the run. Measuring it directly means storing the trajectory, which is expensive. A scalar log costs nothing, and a run already keeps several, so we ask how much of the active dimension survives the projection onto one.

The instrument is standard: delay reconstruction (Takens, 1981; Sauer et al., 1991) followed by the pooled Levina–Bickel estimator (Levina & Bickel, 2004; MacKay & Ghahramani, 2005). Its hazard is that it returns a plausible number when no invariant set exists. Power-law noise yields a finite reproducible dimension set by its spectral exponent (Osborne & Provenzale, 1989), and a decaying

transient, the shape a converging run has, yields a large one. Most of the work below therefore goes into deciding which case one has, beginning with a validation protocol fixed before any result (section 4).

Contributions. We separate four quantities that are frequently identified with one another — the dimension of the subspace the optimiser may move in, the rank of the map from it to the outputs, the number of independent components of the set it visits, and the effective rank of its covariance — and estimate only the third, measuring the fourth beside it everywhere so that the two cannot be conflated (section 3.1). We evaluate that estimate on six systems of increasing realism, ending with a constrained head on a network trained on image data: with no per-system calibration it recovers the active dimension on withheld ranks and seeds to within about one component, up to a ceiling we then locate (section 5). We measure what the absolute value requires, and validate two diagnostics that identify the two failure regimes from the series alone (section 6). Finally we apply all of it to grokking (section 7): the diagnostics refuse both published settings, the trajectory effective rank collapses at generalisation and re-expands, and at a window matched to that transition the estimate on an ordinary log follows it down, in the generalising runs and in neither control (section 7.3).

2 RELATED WORK

Reconstruction and intrinsic dimension. Delay reconstruction rests on Takens’ theorem and its extensions (Takens, 1981; Sauer et al., 1991; Stark, 1999; Stark et al., 2003), is used on empirical series (Sugihara & May, 1990; Sugihara et al., 2012), and takes its lag and dimension from mutual information, false nearest neighbours or singular spectra (Fraser & Swinney, 1986; Kennel et al., 1992; Cao, 1997; Broomhead & King, 1986). We use pooled maximum likelihood (Levina & Bickel, 2004; MacKay & Ghahramani, 2005), not the alternatives (Grassberger & Procaccia, 1983; Facco et al., 2017; Camastra & Staiano, 2016); a finite record bounds them all (Eckmann & Ruelle, 1992). Oversampling (Theiler, 1986) and power-law noise (Osborne & Provenzale, 1989) produce spurious dimensions; surrogate tests (Theiler et al., 1992; Schreiber & Schmitz, 1996) detect nonlinear structure but not its amount, missing the transient regime entirely.

Dimension in optimisation. Random-subspace training (Li et al., 2018; Aghajanyan et al., 2021) reports the smallest k that still solves a task, a property of the task and not of a window; we borrow the parameterisation and measure something else inside it. Nearest to the active dimension is the finding that descent occurs largely in a small subspace spanned by the top Hessian eigenvectors (Gur-Ari et al., 2018; Sagun et al., 2017). The intrinsic dimension of data or of representations (Ansuini et al., 2019; Pope et al., 2021) is a point cloud’s, not a trajectory’s, and the participation ratio we adopt (Gao et al., 2017) is not the spectral-entropy quantity of the same name (Roy & Vetterli, 2007). The one prior measurement of our object we know of is incidental: nearly all of a grokking trajectory’s variance lies in two principal components (Notsawo Jr. et al., 2023). All of these need weights, gradients or activations; the nearest precedent for a scalar computed during training and acted on is the gradient noise scale (McCandlish et al., 2018). Mini-batch descent near a minimum is modelled as a stochastic process (Mandt et al., 2017), our third regime; whether the edge of stability (Cohen et al., 2021) supplies the recurrence the first regime needs, we measure (Appendix Q).

Grokking. Delayed generalisation (Power et al., 2022) is studied in two settings and we use both. The first is regularised and, in our version, stochastic: the transformer configuration of Nanda et al. (2023), also used by Liu et al. (2023), on modular addition and on the S_5 task of Power et al. (2022). The second is the unregularised full-batch gradient descent of Gromov (2023), whose closed form supplies an analytic reference, on the modular polynomials and representability criterion of Doshi et al. (2024), which supply label-matched pairs; Appendix O states where each departs from its source. Explanations are framed in weight or activation space (Nanda et al., 2023; Varma et al., 2023; Liu et al., 2022; Barak et al., 2022) or in a lazy-to-rich transition (Kumar et al., 2024). Closest to us, Notsawo Jr. et al. (2023) predict grokking from early loss oscillations (Thilak et al., 2022) and Merrill et al. (2023) track competing subnetworks. We estimate a dynamical quantity with an independently measured ground truth.

3 PROBLEM STATEMENT

3.1 FOUR QUANTITIES, AND THE ONE WE ESTIMATE

A *window* $\mathcal{W}$ is a block of L consecutive optimiser steps. Everything below is defined on one window, and a run is described by a sequence of them, because what the optimiser is doing changes over training. Let $\theta_t \in \mathbb{R}^P$ be the parameter vector at step t , confined by construction or by the dynamics to an affine subspace $\theta = \theta_0 + V^\top c$ with $V \in \mathbb{R}^{k \times P}$ orthonormal, so that $c_t \in \mathbb{R}^k$ holds the coordinates that can move. Three dimensions attach to $\mathcal{W}$, and they are different numbers.

The *available* dimension is k : how many directions the parameterisation permits. The *functional* dimension is the rank of $\partial f(c)/\partial c$, with f the outputs on a fixed probe set; it bounds how many of those k directions can change the computed function, and we report it rather than estimate it. The *active* dimension is the subject of this paper: how many independent directions the optimiser actually explores inside the window. Let $\mathcal{M}_{\mathcal{W}} = \{c_t : t \in \mathcal{W}\}$ be the set of points the trajectory visits there, and

$$d_{\text{act}}(\mathcal{W}) = \dim_{\varepsilon} \mathcal{M}_{\mathcal{W}} \quad (1)$$

its box-counting dimension at resolution ε : cover $\mathcal{M}_{\mathcal{W}}$ with boxes of side ε , and d_{act} is the exponent d for which the number needed grows like ε^{-d} , so that a curve needs ε^{-1} boxes and a surface ε^{-2} .

Two features of that definition are deliberate. First, ε *stays finite* where the textbook definition would let it go to zero. It has to: the estimator only ever sees the set through distances between nearby samples, so what it can report is the dimension at the scale of its own neighbourhoods. Second, *the window must be long*: any orbit looks like a curve over a short enough stretch, so $d_{\text{act}} = 1$ whatever the motion unless $\mathcal{W}$ outlasts the time the trajectory takes to return near itself.

The case the validation systems build is $c_t = F(\varphi_1(t), \dots, \varphi_r(t))$, with r independent quasiperiodic phases: the orbit closes onto an r -torus and $d_{\text{act}} = r$ at every ε below the weakest mode's amplitude. Above that scale the weakest modes are invisible and d_{act} declines, which is a fact about the set at that resolution rather than an error (section 6.4).

A fourth quantity is measured throughout and is not the active dimension. For a non-negative spectrum $\lambda_1, \dots, \lambda_n$ the participation ratio

$$\text{PR}(\lambda) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \quad (2)$$

equals m exactly when m of the λ_i are equal and the rest vanish. On $\text{spec Cov}\{c_t : t \in \mathcal{W}\}$ it gives the *effective rank* $\text{PR}^{\text{pos}}(\mathcal{W})$, the sense that term carries below, with PR^{det} the same after the per-window linear detrend a drifting trajectory needs and PR^{upd} the same on the increments. It is not that number: it equals r when r directions carry comparable variance and falls below r when they do not, while d_{act} stays at r (section 6.4). We use it to check that a construction excites the r directions it claims, and section 7.2 measures it on a real run. That the four come apart is not hypothetical: in section 5.4 an intervention that leaves the available and functional dimensions and the injected rank untouched still lowers the effective rank (table 15).

3.2 OBSERVATION MODEL AND ESTIMATOR

We observe a scalar series $x_t = \phi(s_t)$, with s_t the full optimiser state and ϕ fixed before the run, and reconstruct $y_t = (x_t, \dots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a generic pair of dynamics and observation on a compact invariant manifold of integer dimension d this map is an embedding once $E > 2d$ (Takens, 1981); the prevalence form replaces d by the box-counting dimension, which need not be an integer, and adds conditions on periodic orbits (Sauer et al., 1991). Genericity is not verifiable for a ϕ fixed in advance, so every result below is reported over a family of observers. We estimate the intrinsic dimension of $\{y_t\}$ by maximum likelihood on nearest-neighbour distances (Levina & Bickel, 2004). Let $r_1^{(i)} \leq \dots \leq r_m^{(i)}$ be the distances from point i to its m nearest neighbours, subject to a Theiler exclusion W_T on their separation in time (Theiler, 1986; Kantz & Schreiber, 2004). Pooling the likelihood over the N points before inverting, as MacKay & Ghahramani (2005) argue one should

Table 1: The set the orbit fills in each regime, the active dimension of equation (1) that follows, and the value the estimator returns on the systems of section 5. The last two rows fail differently: on a transient the estimand exists and the protocol misses it, while under stochastic forcing no invariant measure makes the visited set’s dimension a property of the optimiser at all, so there is nothing to miss.

regime	what the orbit fills	d_{act}	estimate here
deterministic, recurrent	an r -torus	r	r
deterministic, transient	a curve, never revisited	1	≈ 29 , at every r
stochastically driven	no low-dimensional invariant set	none	set by E_{max} , not by r

since the reciprocal is the unbiased quantity, gives

$$\hat{d}_{\text{MG}}^{-1} = \frac{1}{N(m-1)} \sum_{i=1}^N \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}, \quad (3)$$

while averaging the per-point estimates instead, as Levina & Bickel (2004) propose, gives the variant we write LB. We call what MG returns *the estimate* and the procedure *the estimator*; table 9 reports both and Appendix A gives the algorithm as implemented. Three further statistics accompany every estimate, each naming an effect that can produce a dimension signal with no geometry: the *roughness* $\text{std}(\Delta x)/\text{std}(x)$, the *linear participation ratio* PR_{delay} of equation (2) on $\text{Cov}\{\mathbf{y}_t\}$, and the *spectral participation ratio* on the power spectrum of x .

3.3 DYNAMICAL REGIMES

Two conditions must hold, and they are not the same. The theorems require a deterministic flow on a compact invariant set, which licenses the delay map as an embedding. The neighbour statistic needs more: the record must visit that set densely enough for a point’s nearest neighbours to be returns rather than its own temporal neighbours, which the Theiler exclusion enforces and section 3.4 tests. Recurrence enters here and only here — not as a requirement of the estimator, which would accept points drawn from independent runs, but as one of a protocol whose whole sample is a single trajectory. Neither condition is needed for d_{act} to be the optimiser’s; both are needed to read it off one scalar record. An optimiser near a solution may be in any of the three regimes of table 1, and only the first meets them.

Mini-batch gradient noise produces the third regime, and theory serves it worse than it looks. The embedding theorems for forced systems (Stark, 1999; Stark et al., 2003) cover it in part: the update may depend on the state arbitrarily, and their iterated-function-system form admits a driver selecting among finitely many maps, as sampling a finite dataset does. But they deliver an embedding *fibre by fibre*, one per realisation of the drive, and a single run crosses fibres, so no invariant measure exists there to sample. Coloured noise alone yields a finite reproducible dimension set by its spectral exponent (Osborne & Provenzale, 1989), or at least an estimate that behaves like one (Theiler, 1991). The second regime is less often noticed: a decaying transient traces a curve it never returns to, so it has no $\mathcal{M}_{\mathcal{W}}$, and its effective rank remains near unity (table 15) only because that curve is nearly straight.

3.4 ADMISSIBILITY DIAGNOSTICS

Two statistics computed from the series alone separate the two failure regimes. The **identifiability ratio** $\rho_{\text{ident}} = \hat{d}_{\text{MG}}(2E_{\text{max}})/\hat{d}_{\text{MG}}(E_{\text{max}})$ is near unity when a dimension is resolvable and grows when the estimate is a property of the embedding space rather than of the data, which is the stochastic regime. On a transient $\rho_{\text{ident}} \approx 1$ as well, for the wrong reason, and the **trend-crossing count** distinguishes the two: the number of sign changes of the window’s residual about its least-squares line, which is near zero there. It tests non-monotonicity of the series, not recurrence in the reconstructed space, so the pair must be read together. A third statistic, the **degeneracy indicator**, tests no regime: it is raised when too many neighbour distances or log-ratio sums reach their numerical floors, and guards against quantised or constant observers (Appendix A). Reference values are in section 6.4.

4 VALIDATION PROTOCOL

Six requirements were fixed before any result, and each is named so that table 5 can say which a system fails.

- (1) *Constructed truth*. Every system is built with r independent quasiperiodic phases, so $d_{\text{act}} = r$ by construction, and the recorded trajectory’s effective rank confirms that all r are comparably excited.
- (2) *Frozen configuration*. The estimator is chosen once, by a search over estimator settings only. Searching the system’s settings at the same time — the drive period, say — would choose the data along with the method, and the winning cell would report how favourable a system we found rather than how well the estimator works.
- (3) *Disjoint split*, in seed and in rank. Withholding seeds alone is not enough: whenever the excitation geometry is a function of r , every seed at a given rank shares it.
- (4) *Silence control*. Re-run each system with the learning rate set to zero. The parameters then never move, so an observer of the optimiser must be flat: a constant log is exactly the right answer here. If instead the log keeps varying and returns the same estimate as before, the observer was never reading the optimiser. That happened in two of our systems: the drive acts on the targets, so the loss continues to move through the residual while the parameters are fixed.
- (5) *Companion statistics*. The roughness, the linear participation ratio and the spectral participation ratio of section 3.2 are reported with every estimate, and where one of them does as well as the estimator we say so.
- (6) *Matched bandwidth* across ranks. In a construction where each added direction also adds a higher frequency, the roughness alone orders the rank, and no embedding is being tested.

5 EVALUATION ON SYSTEMS WITH A KNOWN ACTIVE DIMENSION

Six systems of increasing realism are used, and table 5 collects the outcome: an oscillating diagonal matrix, online linear and logistic regression, a frozen nonlinear decoder, a perceptron in a k -subspace, and a linear head on a network trained on image data. In each the orbit closes onto an r -torus by construction, the effective rank confirms it, and the estimator sees one scalar series.

5.1 AN OSCILLATING DIAGONAL MATRIX

The first system has no optimiser and no feedback, so a negative result there would be decisive. In $\mathbf{W}(t) = \text{diag}(b_i + \delta_i(t))$, r coordinates oscillate at rationally independent frequencies over non-zero offsets, and the estimate tracks r across the whole range on both held-out seeds (table 5). It also supplies the geometry control. Drive several coordinates from a single phase and the set they trace has one degree of freedom rather than several, though it is no smoother for it; the estimate separates that case from the same number of independent oscillators several-fold, while the roughness of the series distinguishes them by a few per cent. Smoothness alone cannot produce that separation.

5.2 THE CEILING

The estimator saturates above about eight directions. Sweeping the embedding dimension against the record length separates two things we had been treating as one (Appendix R). The *level* returned at high rank belongs to the embedding: widening the delay window raises it, lengthening the record does not. The *rank up to which the estimate still tracks the truth* belongs to neither knob — it never passes about ten components, and reaches that at the *shortest* record examined.

Both candidate explanations therefore fail: a finite-record bound (Eckmann & Ruelle, 1992) would allow far fewer components than we recover, and $E > 2d$, right in direction, is wrong in form, the ceiling climbing with E_{max} an order of magnitude more slowly than that rule requires, so $E_{\text{max}}/2$ matched only at the value we happened to freeze. One caution is substantial: a purely linear statistic of the delay cloud follows the same curve, so much of what saturates may be how many components the delay window resolves linearly, not geometry.

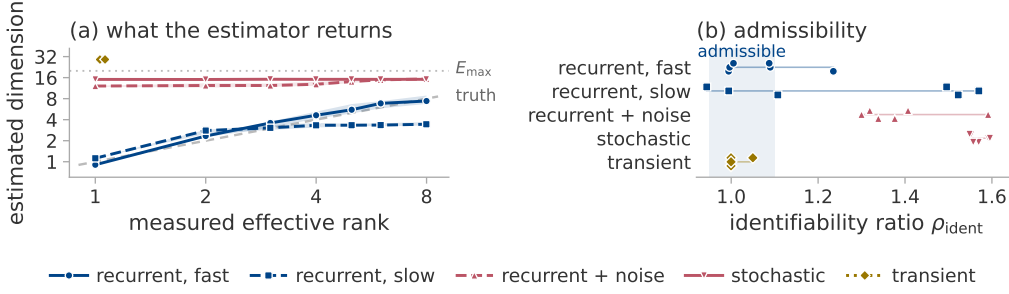


Figure 1: Whether the estimate recovers the active dimension, and whether the diagnostics separate the regimes with no ground truth. The evaluation of section 5.4: a frozen tanh network on the digits data, a linear head confined to ten directions of which r are excited, five of the nine excitation cases; colour is the regime of table 1. (a) The estimate against the measured effective rank, which the construction holds at r ; the marker is the median of forty values, four seeds by ten observers, and the band their interquartile range. The dashed diagonal is the truth and the dotted line $E_{\max}$, a reference and not a bound: no clamping is applied, and the transient case lies above it, its seven points piled at $x \approx 1.05$ because its measured rank is about one at every r . (b) The identifiability ratio on a smaller grid — seed 0, three observers, $r \in \{2, 6\}$. The transient is the one regime it passes, hence the trend-crossing count of section 3.4.

5.3 THE SILENCE CONTROL

Requirement 4 asks that a zero learning rate silence the observer; most published analyses of dimension on training logs do not run it. It invalidated two of our six systems, which read naively succeed yet reproduce their reported results with the parameters frozen. It also rejects one observer, the instantaneous mini-batch loss of section 5.4, whose estimate goes on tracking the nominal rank when the parameters cannot move, though it is otherwise among the more accurate observers. Two further systems fail requirement 2 (table 5, Appendix D).

5.4 A CONSTRAINED HEAD ON A NETWORK TRAINED ON IMAGE DATA

The sixth system uses real inputs: a tanh network trained on the scikit-learn handwritten digits and frozen, with a linear head confined to a k -dimensional affine subspace under softmax cross-entropy, so the curvature is data-dependent and all the dynamics occur in c . The excitation is made r -dimensional by modulating the loss weights of r of twelve fixed data groups and orthogonalising the resulting gradient directions (Appendix F); the measured effective rank then matches r (table 15). Nine excitation cases are run, spanning the three regimes of table 1, and figure 1 shows five of them.

Twelve observers are recorded (Appendix B), and the estimator’s parameters were frozen on four of them and on reserved seeds and ranks, so what follows is out-of-sample in rank and seed throughout and in observer for most of what is scored. In the deterministic recurrent regime the estimate recovers the active dimension to within about one component, and every observer orders the withheld ranks correctly (table 5; Appendix D scores them one by one). One caveat attaches to the observers a run already records: the two parameter norms are also the only observers whose roughness by itself orders the withheld ranks, so for them a smoothness explanation cannot be excluded. Twenty available directions give the same picture with a lower ceiling; that configuration is exploratory, having no seed split, a Theiler exclusion smaller than its own embedding span, and $E = 2d$ exactly, one short of section 3.2’s condition (Appendix N).

6 CONDITIONS OF VALIDITY

6.1 DYNAMICAL REGIME

Figure 1 evaluates the estimator in each regime, at one nominal rank, observer and matched octave. Under stochastic excitation the estimate is flat in the rank and well above the true value, responding

to $E_{\max}$ and the record rather than to anything in the data. Weak noise on an otherwise recurrent trajectory is less damaging: it destroys the absolute value and leaves the ordering.

A decaying transient returns near no state it has left, so the set it visits is a curve and $d_{\text{act}} = 1$. The estimate exceeds that by more than an order of magnitude (table 1), and the Theiler exclusion is why. Remove it and the estimator returns, at every rank, seed and observer, the value maximum likelihood must return on a uniformly sampled curve, while the recurrent cases do not move (Appendix P). Nor is the value a level: it is a divergent quantity read at the exclusion cap, and that it exceeds $E_{\max}$ is the expected behaviour of a maximum likelihood fit whose model is violated rather than a bound failing. No clamping is applied (Appendix A).

Every system here is driven and training is not, so we test the one undriven candidate the literature offers: the edge of stability (Cohen et al., 2021). The perceptron of section 7 reaches it, its sharpness self-limiting while the loss rises on half of all steps. But oscillation is not recurrence, and here the diagnostics invert the classification: they admit the runs that never return near a state they have left, and refuse the minority that do (Appendix Q).

6.2 THE DELAY LAG

The delay window spans $(E_{\max} - 1)\tau$ samples, and it must cover a real fraction of the oscillation period; otherwise the delay coordinates are collinear and the torus never unfolds. That the window rather than the lag governs this is standard (Casdagli et al., 1991). On a fixed torus, varying τ alone moves the estimate over more than an order of magnitude, and it passes through the truth only where the delay window covers a few tenths of a period (table 11, figure 4).

The right lag follows from the oscillation period, which a controlled experiment knows and a training log does not, and we do not solve this. The lag used in section 7 is the frozen one, transplanted from a calibration system whose drive period is far shorter than the autocorrelation time of those logs, which by itself forbids reading an absolute value there. The identifiability ratio inherits the problem: doubling $E_{\max}$ doubles the delay span, so a large value means either that the estimate belongs to the embedding space or that the lag is mismatched, and on the grokking logs both readings are live.

6.3 NUISANCE FACTORS

At a fixed active dimension of four we vary seven factors that leave it unchanged, over five seeds and twelve observers, and read each against a real four-component change (table 12). Two matter. Smoothness passes on most observers: shifting the drive band by an octave changes the roughness sharply and moves the estimate no further than seed-to-seed scatter does, though it fails on the two parameter norms, whose roughness orders the withheld ranks perfectly. The second factor is the observer's gain. A constant rescaling leaves the estimate unchanged, the scorer standardising it away, but a *ramped* gain moves it by about a component — so a run whose parameter norm grows will drift by that much with nothing dimensional occurring. That is the nuisance that matters most in section 7.

6.4 THE ESTIMAND IS A DIMENSION, NOT AN EFFECTIVE RANK

Every construction in section 5 equalises the drive amplitudes, which makes the active dimension and the effective rank of section 3.1 numerically equal, and an experiment in which two candidate estimands agree cannot say which the estimator reports. Driving mode l at amplitude q^l separates the two: the orbit still closes onto an r -torus, so $d_{\text{act}} = r$, while the effective rank decays with q .

The estimate follows the dimension (table 14, figure 5): as the drive is made anisotropic the effective rank decays while the estimate stays with the number of modes, and across the grid it is closer to that number than to the effective rank. It registers only the modes it can resolve, though: once the weakest lies far enough below the strongest, the estimate decays with the rank, the torus it can see having fewer dimensions than the one built.

The diagnostics. With no ground truth, the identifiability ratio separates the accurate cases from the inaccurate ones without overlap, within a regime as well as between them. It is not sufficient alone — on a transient $\rho_{\text{ident}} \approx 1$ as well, for the wrong reason — and must be paired with the trend-crossing

count, which is near zero there. Both are calibrated on these five cases only, which is why section 7 treats the admissible band as a description and not a decision rule.

7 APPLICATION TO GROKING

We work in both settings of section 2, which differ in every relevant respect (Appendix O lists every run and every departure from the sources). The transformer runs are regularised, trained with AdamW under cross-entropy on mini-batches where the source is full-batch, which puts them in the stochastic regime of table 1. The perceptron runs are unregularised full-batch descent on mean squared error; we take the tasks and representability criterion of Doshi et al. (2024) but not their optimiser (Appendix M). The diagnostics refuse the level both settings’ logs report, so we pay for the trajectory instead, and then return to the logs with the one question a refused level still permits.

7.1 THE DIAGNOSTICS REFUSE BOTH SETTINGS

The frozen configuration of section 5.4, with only the window and stride shortened to fit these records, gives table 16 and figure 6. The two settings fail in different corners, and neither failure appeals to any label. The regularised mini-batch runs occupy a band of the identifiability ratio the deterministic regimes never reach, and their estimates scatter widely across seeds of one configuration, set by $E_{\max}$ and the record rather than by any rank, exactly as figure 1 predicts for stochastic excitation. The full-batch runs carry a transient’s signature instead: $ho_{\text{ident}} \approx 1$ with almost no trend crossings. In neither setting may the value be read as a dimension.

A change inside a run might still be read, since the offset a fixed configuration imposes is constant within a run, and the floor is the drift a ramped gain alone produces (section 6.3). Not at this window: it spans tens of thousands of optimiser steps against a transition hundreds wide, a want of resolution and not of signal, so section 7.3 shortens it (Appendix L).

Label-matched pairs. The perceptron setting supplies the control the first cannot: both members of a pair fit the training set completely, under identical optimisation, and differ by one monomial. The estimate comes out higher for the non-generalising member in every pair, with no overlap between the groups — and it is still not a dimension. All eight runs are transient, the separation is already visible in the raw loss curves, the direct measurement does not reproduce it, and a fall is not specific to generalisation: one occurs in a long run that never leaves chance accuracy (Appendices G, J and K).

7.2 THE TRAJECTORY SIMPLIFIES AT GENERALISATION

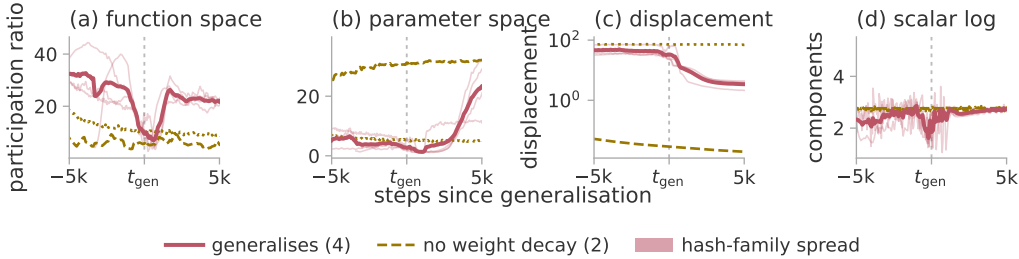


Figure 2: Whether the trajectory simplifies at generalisation, and whether the controls do the same. Three regularised seeds of modular addition and one S_5 composition, against the same two configurations without weight decay, which memorise and stay at chance accuracy. Colour is the regime of table 1; the thick line is the mean of the four runs that generalise, and each control is aligned on the generalisation step of its matched run. The statistic in (a)–(c) is PR^{det} , an effective rank: none of these runs recurs, so none supports reading a dimension. Windows are 600 optimiser steps, 300 for S_5 , plotted at their midpoint; the band on (a) and (b) is the two-hash disagreement (Appendix I). Panel (d) is a different measurement on the same midpoints: the delay estimate of section 7.3 on the parameter-norm log.

A stored trajectory delivers what a log cannot: the effective rank of section 3.1 is defined whether or not the orbit recurs. We sketch it in parameter and in function space at every logged step of a transformer trained to grok (Appendix I). Near generalisation it collapses (figure 2): in function space by close to an order of magnitude, in parameter space to near unity, each within a few hundred steps of the transition, and both then re-expand (table 17). Two things support the reading. The collapse occurs at generalisation in seeds whose generalisation steps differ several-fold, and in a second task, so it is not tracking training progress; and the displacement reaches its floor thousands of steps later and never recovers, where the rank does, so the trajectory has not simply stopped moving. The effective rank measures the variation inside the window, not the extent of the set the orbit visits, so the collapse is in that variation and not in d_{act} .

The controls. In the transition window the controls decline much less than the generalising runs, but over their whole budget one of them declines just as far (table 17): depth does not separate the groups, timing and shape do. Both controls peak early and then settle, and their parameter norm never returns to its earlier level, where every generalising run gives back much of its growth. A low rank is reached both by a run that has come to rest and by one about to generalise, and that growing norm is the ramped gain of section 6.3, the mechanism behind the scalar-log decline of section 7.1.

7.3 THE LOG ESTIMATE FALLS WHERE THE TRAJECTORY DOES

Setting the window in units of the transition instead of the record makes the test of section 7.1 possible. We re-run the estimator on the same logs, at the window and midpoints the direct measurement uses, so the two statistics are sampled at the same instants. The frozen configuration will not fit in a window that short, and requirement 2 forbids choosing a replacement on the outcome, so we report every cell of a grid and take as headline the one a rule blind to the outcome selects (Appendix L).

On the parameter norm the estimate halves at the generalisation step in all four generalising runs, with its floor on the same side of the transition as the direct measurement's (figure 2d), while the controls show no comparable movement. That drop is among the largest each generalising run ever shows and unremarkable in the controls, and all but one of the usable grid cells separate the groups (table 20). Nor is it the observer's shape, the nuisance section 6.3 identifies: against surrogates that keep the log's shape and its fluctuation spectrum, every generalising run's p lies below every control's, at every smoothing length and seeding tried (Appendix L).

Two signatures of one event, then, one paid for and one already written by the run. Three things keep the second modest. Its level cannot be read as a dimension, the identifiability ratio being uncomputable here, so only the change is read. The observer is the parameter norm, for which section 6.3 cannot exclude a smoothness explanation, so the surrogate control and not the level carries the claim. And a simpler statistic registers it more strongly: the roughness of the same windows collapses by an order of magnitude where the estimate merely halves, so most of what this log carries about the transition is that collapse, which needs no embedding at all.

8 CONCLUSION

In a deterministic recurrent system with a matched delay lag, one scalar log of a suitable observer counts the independent components of the set the optimiser visits, to within about one component and up to about eight. Outside that regime the number it returns is not that dimension: a transient's visited set is a curve, and stochastic forcing leaves no invariant set to have one. The instrument is therefore usable only with the pair of diagnostics beside it, and on grokking they refuse both standard settings. Measured directly instead, the trajectory effective rank collapses at generalisation and re-expands, much less in the controls, and at a matched window the log estimate follows it down.

Limitations. The measurement needs a recurrent regime and a lag matched to a period no log reveals; above about eight components it is only ordinal, and the configuration is a choice as much as a measurement (Appendix C). The collapse rests on four runs against two controls differing in the variable that also drives the parameter norm; it is a signature, not an early warning, and does not reproduce in the second setting. The matched-window drop is close to what a ramped gain alone produces, one control has little range to move through, and the edge-of-stability evidence is seven runs on one task (Appendices J, L and Q).

REFERENCES

- Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 7319–7328, 2021.
- Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. Intrinsic dimension of data representations in deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Boaz Barak, Benjamin L. Edelman, Surbhi Goel, Sham Kakade, Eran Malach, and Cyril Zhang. Hidden progress in deep learning: SGD learns parities near the computational limit. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- David S Broomhead and Gregory P King. Extracting qualitative dynamics from experimental data. *Physica D: Nonlinear Phenomena*, 20(2-3):217–236, 1986.
- Francesco Camastra and Antonino Staiano. Intrinsic dimension estimation: Advances and open problems. *Information Sciences*, 328:26–41, 2016. doi: 10.1016/j.ins.2015.08.029.
- Liangyue Cao. Practical method for determining the minimum embedding dimension of a scalar time series. *Physica D: Nonlinear Phenomena*, 110(1-2):43–50, 1997. doi: 10.1016/S0167-2789(97)00118-8.
- Martin Casdagli, Stephen Eubank, J. Doyne Farmer, and John Gibson. State space reconstruction in the presence of noise. *Physica D: Nonlinear Phenomena*, 51(1-3):52–98, 1991. doi: 10.1016/0167-2789(91)90222-U.
- Moses Charikar, Kevin Chen, and Martin Farach-Colton. Finding frequent items in data streams. In *International Colloquium on Automata, Languages, and Programming*, pp. 693–703, 2002.
- Kenneth L. Clarkson and David P. Woodruff. Low rank approximation and regression in input sparsity time. In *Proceedings of the 45th Annual ACM Symposium on Theory of Computing (STOC)*, pp. 81–90, 2013. doi: 10.1145/2488608.2488620.
- Jeremy M Cohen, Simran Kaur, Yuanzhi Li, J Zico Kolter, and Ameet Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. In *International Conference on Learning Representations (ICLR)*, 2021. URL <https://openreview.net/forum?id=jh-rTtvkGeM>.
- Darshil Doshi, Tianyu He, Aritra Das, and Andrey Gromov. Grokking modular polynomials. *arXiv preprint arXiv:2406.03495*, 2024.
- Jean-Pierre Eckmann and David Ruelle. Fundamental limitations for estimating dimensions and lyapunov exponents in dynamical systems. *Physica D: Nonlinear Phenomena*, 56(2-3):185–187, 1992. doi: 10.1016/0167-2789(92)90023-G.
- Elena Facco, Maria d’Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7(1):12140, 2017.
- Andrew M Fraser and Harry L Swinney. Independent coordinates for strange attractors from mutual information. *Physical review A*, 33(2):1134, 1986. doi: 10.1103/PhysRevA.33.1134.
- Peiran Gao, Eric Trautmann, Byron M. Yu, Gopal Santhanam, Stephen Ryu, Krishna Shenoy, and Surya Ganguli. A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv* 214262, 2017. doi: 10.1101/214262.
- Peter Grassberger and Itamar Procaccia. Measuring the strangeness of strange attractors. *Physica D: Nonlinear Phenomena*, 9(1-2):189–208, 1983.
- Andrey Gromov. Grokking modular arithmetic. *arXiv preprint arXiv:2301.02679*, 2023.
- Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. *arXiv preprint arXiv:1812.04754*, 2018.

- Holger Kantz and Thomas Schreiber. *Nonlinear Time Series Analysis*. Cambridge university press, 2004.
- Matthew B Kennel, Reggie Brown, and Henry DI Abarbanel. Determining embedding dimension for phase-space reconstruction using a geometrical construction. *Physical review A*, 45(6):3403, 1992. doi: 10.1103/PhysRevA.45.3403.
- Tanishq Kumar, Blake Bordelon, Samuel J. Gershman, and Cengiz Pehlevan. Grokking as the transition from lazy to rich training dynamics. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=vt5mnLVIVo>.
- Elizaveta Levina and Peter J. Bickel. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 17, 2004. URL <https://proceedings.neurips.cc/paper/2004/file/74934548253bcab8490ebd74afecebb5-Paper.pdf>.
- Chunyuan Li, Heerad Farkhoor, Rosanne Liu, and Jason Yosinski. Measuring the intrinsic dimension of objective landscapes. In *International Conference on Learning Representations (ICLR)*, 2018. URL <https://openreview.net/forum?id=ryup8-WCW>.
- Ziming Liu, Ouail Kitouni, Niklas Nolte, Eric J. Michaud, Max Tegmark, and Mike Williams. Towards understanding grokking: An effective theory of representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Ziming Liu, Eric J. Michaud, and Max Tegmark. Omnigrok: Grokking beyond algorithmic data. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/forum?id=zDiHoIWa0q1>.
- David J C MacKay and Zoubin Ghahramani. Comments on ‘maximum likelihood estimation of intrinsic dimension’ by e. levina and p. bickel, 2005. URL <http://www.inference.org.uk/mackay/dimension/>. Unpublished note.
- Stephan Mandt, Matthew D Hoffman, and David M Blei. Stochastic gradient descent as approximate Bayesian inference. *Journal of Machine Learning Research*, 18(1):4873–4907, 2017.
- Sam McCandlish, Jared Kaplan, Dario Amodei, and OpenAI Dota Team. An empirical model of large-batch training. *arXiv preprint arXiv:1812.06162*, 2018.
- William Merrill, Nikolaos Tsilivis, and Aman Shukla. A tale of two circuits: grokking as competition of sparse and dense subnetworks. *arXiv preprint arXiv:2303.11873*, 2023.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023.
- Pascal Tikeng Notsawo Jr., Hattie Zhou, Mohammad Pezeshki, Irina Rish, and Guillaume Dumas. Predicting grokking long before it happens: A look into the loss landscape of models which grok. *arXiv preprint arXiv:2306.13253*, 2023.
- A. R. Osborne and A. Provenzale. Finite correlation dimension for stochastic systems with power-law spectra. *Physica D: Nonlinear Phenomena*, 35(3):357–381, 1989.
- Phil Pope, Chen Zhu, Ahmed Abdelkader, Micah Goldblum, and Tom Goldstein. The intrinsic dimension of images and its impact on learning. In *International Conference on Learning Representations (ICLR)*, 2021. URL <https://openreview.net/forum?id=XJk19XzGq2J>.
- Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- A. Provenzale, L. A. Smith, R. Vio, and G. Murante. Distinguishing between low-dimensional dynamics and randomness in measured time series. *Physica D: Nonlinear Phenomena*, 58(1-4): 31–49, 1992.

- Olivier Roy and Martin Vetterli. The effective rank: a measure of effective dimensionality. In *European Signal Processing Conference (EUSIPCO)*, pp. 606–610, 2007.
- Levent Sagun, Utku Evci, V. Uğur Güney, Yann Dauphin, and Léon Bottou. Empirical analysis of the Hessian of over-parametrized neural networks. *arXiv preprint arXiv:1706.04454*, 2017.
- Tim Sauer and James A. Yorke. How many delay coordinates do you need? *International Journal of Bifurcation and Chaos*, 3(3):737–744, 1993.
- Tim Sauer, James A. Yorke, and Martin Casdagli. Embedology. *Journal of Statistical Physics*, 65(3-4):579–616, 1991.
- Thomas Schreiber and Andreas Schmitz. Improved surrogate data for nonlinearity tests. *Physical Review Letters*, 77(4):635–638, 1996.
- Jaroslav Stark. Delay embeddings for forced systems. I. Deterministic forcing. *Journal of Nonlinear Science*, 9(3):255–332, 1999. doi: 10.1007/s003329900072.
- Jaroslav Stark, David S Broomhead, ME Davies, and J Huke. Delay embeddings for forced systems. II. Stochastic forcing. *Journal of Nonlinear Science*, 13(6):519–577, 2003. doi: 10.1007/s00332-003-0534-4.
- George Sugihara and Robert M May. Nonlinear forecasting as a way of distinguishing chaos from measurement error in time series. *Nature*, 344(6268):734–741, 1990. doi: 10.1038/344734a0.
- George Sugihara, Robert May, Hao Ye, Chih-hao Hsieh, Ethan Deyle, Michael Fogarty, and Stephan Munch. Detecting causality in complex ecosystems. *Science*, 338(6106):496–500, 2012. doi: 10.1126/science.1227079.
- Floris Takens. Detecting strange attractors in turbulence. In *Dynamical systems and turbulence, Warwick 1980*, pp. 366–381. Springer, 1981. doi: 10.1007/BFb0091924.
- James Theiler. Spurious dimension from correlation algorithms applied to limited time-series data. *Physical Review A*, 34(3):2427–2432, 1986.
- James Theiler. Some comments on the correlation dimension of $1/f^\alpha$ noise. *Physics Letters A*, 155(8-9):480–493, 1991.
- James Theiler, Stephen Eubank, André Longtin, Bryan Galdrikian, and J. Doyne Farmer. Testing for nonlinearity in time series: the method of surrogate data. *Physica D: Nonlinear Phenomena*, 58(1-4):77–94, 1992.
- Vimal Thilak, Etai Littwin, Shuangfei Zhai, Omid Saremi, Roni Paiss, and Joshua M Susskind. The slingshot mechanism: An empirical study of adaptive optimizers and the loss landscape. *arXiv preprint arXiv:2206.04817*, 2022. URL <https://arxiv.org/abs/2206.04817>.
- Vikrant Varma, Rohin Shah, Zachary Kenton, János Kramár, and Ramana Kumar. Explaining grokking through circuit efficiency. *arXiv preprint arXiv:2309.02390*, 2023.
- Kilian Weinberger, Anirban Dasgupta, John Langford, Alex Smola, and Josh Attenberg. Feature hashing for large scale multitask learning. In *International Conference on Machine Learning (ICML)*, pp. 1113–1120, 2009.

A THE ESTIMATOR

Algorithm 1 is the estimator as it was run. It computes equation (3) with three departures that affect what is reported, and it fixes the constants the main text leaves unnamed.

Algorithm 1 Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

Require: series $x_{1:n}$; lag τ ; embedding dimension E ; neighbours m ; Theiler exclusion W_T ; dither σ

- 1: $z_t \leftarrow (x_t - \text{mean } x) / \text{std } x + \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$ ▷ breaks exact ties
- 2: $\mathbf{y}_t \leftarrow (z_t, z_{t-\tau}, \dots, z_{t-(E-1)\tau})$ for $t = (E-1)\tau + 1, \dots, n$
- 3: $S \leftarrow 0; N \leftarrow 0; n_{\text{dist}} \leftarrow 0; n_{\text{ratio}} \leftarrow 0$
- 4: **for each** $\mathbf{y}_i$ **do**
- 5: $r_1^{(i)} \leq \dots \leq r_m^{(i)} \leftarrow$ distances to the m nearest $\mathbf{y}_j$ with $|i - j| > W_T$
- 6: $r_j^{(i)} \leftarrow \max(r_j^{(i)}, \varepsilon_{\text{dist}})$ for each j ; count those clipped into n_{dist}
- 7: $S_i \leftarrow \max(\sum_{j=1}^{m-1} \log(r_m^{(i)} / r_j^{(i)}), \varepsilon_{\text{ratio}})$; count clips into n_{ratio}
- 8: $S \leftarrow S + S_i; \quad N \leftarrow N + 1$
- 9: **end for**
- 10: mark the window degenerate if n_{dist} or n_{ratio} exceeds 1 % of its denominator
- 11: **return** $(N(m-1) - 1) / S$

The three departures are these. *The returned value is $(N(m-1) - 1) / S$ rather than $N(m-1) / S$* , the exactly unbiased pooled maximum likelihood estimate for a $\Gamma(N(m-1), 1/d)$ sample, which differs from equation (3) by less than 10^{-5} at the sizes used here. *A window marked degenerate is returned rather than discarded*: the ladder pipelines drop marked windows when summarising a series over its sliding windows, the two grokking pipelines take the median over all of them, and no published cell of table 16 contains one. *The returned value is never clamped to E* , so a divergent estimate is reported as divergent; clamping would convert the transient regime of table 1 into a plausible number.

The floors are $\varepsilon_{\text{dist}} = 10^{-8}$ on a neighbour distance and $\varepsilon_{\text{ratio}} = 10^{-5}$ on the per-point log-ratio sum, with $\sigma = 10^{-9}$ on the standardised series. The Theiler exclusion is $W_T = \max((E-1)\tau, t_{\text{acf}})$, where t_{acf} is the first lag at which the autocorrelation falls below $1/e$, capped at 150 samples because the neighbour query cost is linear in it. Appendix N measures what that cap costs.

B THE OBSERVERS

Twelve scalar observers are recorded on the system of section 5.4, in five families; table 2 defines each. Two are excluded from every aggregate, for the reasons appendix D gives. “Parameter norms (two)” in table 7 is the pair $\|\boldsymbol{\theta}\|$ and $\|\mathbf{c}\|$, reported as one row because their errors agree to two decimals; they are distinct quantities and both are given here.

Table 2: The twelve observers. $\mathbf{c}$ is the coordinate in the k -dimensional subspace and $\boldsymbol{\theta}_0$ the frozen head at initialisation; $\mathbf{L}$ is the matrix of probe logits, the probe being a fixed held-out set drawn once before training; $\mathbf{g} = \partial \ell / \partial \mathbf{c}$. The projection directions $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{a}$, and the rotation $\mathbf{R}$, are drawn once from a fixed seed and never redrawn.

name	family	definition
instantaneous loss	loss	the loss on the current minibatch
full-batch loss	loss	the loss on the whole training set
probe loss	loss	cross-entropy on the probe set
parameter norm	norm	$(\ \boldsymbol{\theta}_0\ ^2 + \ \mathbf{c}\ ^2)^{1/2}$
subspace norm	norm	$\ \mathbf{c}\ _2$
function-space norm	norm	$\ \mathbf{L}\ _F$
gradient norm	gradient	$\ \mathbf{g}\ _2$
gradient projection	gradient	$\langle \mathbf{u}, \mathbf{g} \rangle$
fixed parameter projection	projection	$\langle \mathbf{v}, \mathbf{R}\mathbf{c} \rangle$
function-space projection	function	$\langle \mathbf{a}, \text{vec } \mathbf{L} \rangle$
margin	function	mean probe logit gap, true class less best other
probe accuracy	function	accuracy on the probe set

C FROZEN CONFIGURATIONS

Two estimator configurations are used in this paper, one for each range of the active dimension. Both were selected once, on data withheld from every later experiment, and then frozen. Table 3 states them and the grids they came from.

Table 3: The two frozen configurations and their selection. Only the swept parameters were varied; the remainder were fixed before any estimate was computed. Three settings are common to both and are not repeated: $m = 20$ neighbours, the autocorrelation Theiler rule of appendix A, and a dither of 10^{-9} ; the first two were swept in the eight-direction grid and these are the values it selected.

	eight-direction	twenty-direction
$E_{\max}$	20	40
τ	4	16
window / stride	8 000 / 2 000	8 000 / 4 000
swept	$E_{\max}, \tau, m, \text{Theiler, window}$	$E_{\max}, \tau$
grid size	48	4
selection seeds	90, 91, 92	0, 1, 2
selection ranks	2, 4, 6	2, 6, 10, 14, 18
withheld seeds	0–3	none
withheld ranks	1, 3, 5, 8	the other fifteen
selection error, best	0.32	2.47
selection error, worst	1.50	3.35

The selection objective in both cases is the error of the median estimate against the median measured effective rank, over the four observers reserved for selection, one from each observer family. The eight-direction objective adds a penalty on the rank correlation and on the spread across seeds; at the twenty-direction configuration both penalties are identically zero, so the objective there reduces to the error alone.

Five properties of these choices should be stated with them.

Both lie at a grid boundary. The eight-direction configuration is at the top of all five swept axes and the twenty-direction one at the top of both. That is less serious than it appears for the first: four of its five axes have only two levels, so every configuration in that grid is at a boundary in four of them, and τ , the only axis with an interior value, has its winner at the top end. But neither configuration is interior, and a wider grid could move the level.

The selection error spans a factor of five on identical data. Across the eight-direction grid the error runs from 0.32 to 1.50 on the same logs, so the level is a choice as much as a measurement, and only the withheld numbers of section 5.4 may be read as recovery.

The Theiler axis is inert on this data. The autocorrelation rule returns the larger of the embedding span $(E_{\max} - 1)\tau$ and the measured autocorrelation time, and on these logs the autocorrelation time is one to three samples against an embedding span of nine or more. The two Theiler settings therefore return bit-identical values wherever they are compared, and the 48-point grid contains 24 distinct measurements.

Neither configuration always uses the exclusion its parameters name. The implementation caps the Theiler exclusion at 150 samples, which binds unconditionally at the twenty-direction configuration and on five of the seven training logs of section 7. Appendix N states the departure and measures what it costs.

The stride is not always the frozen one. Three pipelines override the window geometry and no estimator field, and table 4 states what each of them uses. The constructed-system setting is the one behind the 0.87 of section 5.4, so “at the frozen configuration” is true there of the estimator and not of the stride. Every other field — $E_{\max}, \tau, m$, the Theiler rule, the dither — is used as frozen throughout, so what moves between experiments is how the record is cut up, not how a window is scored.

Table 4: Which window and stride each pipeline uses, and where they depart from the frozen configuration of table 3. Lengths are in samples of the log. The longer stride on the constructed systems keeps the number of windows near ten whatever window length was frozen, and the second value applies where the absent burn-in makes the record longer. A third of a grokking record is 4 000 samples, which at the logging stride is the 39 990-step window of appendix L.

pipeline	window	stride
frozen configuration	8 000	2 000
constructed systems (section 5)	8 000	3 000, or 3 666
training logs (section 7)	a third of the record	1 000

D PER-OBSERVER RESULTS AND ALTERNATIVE STATISTICS

Table 5 is the ladder of section 5 in full, table 7 scores each observer on its own, and table 9 sets the estimator against every statistic we could compute on the same data. Two rows of the ladder carry a caveat, and they are taken first.

Table 5: How accurately the estimator recovers the active dimension on each system, and which requirements of section 4 that system fails. Systems are ordered by increasing realism. The unit of analysis is one (rank, seed, observer) run, each side of the error being the median over that run’s sliding windows, except the last row, which has one window per run; ρ is over ranks after a median over seeds. The two oscillating-matrix figures are its two held-out seeds.

system	truth	range	MAE	ρ	protocol
oscillating matrix	constructed	1–10	0.30 / 0.69	0.99 / 1.00	fails 2, 5
online linear regression	constructed	1–20	1.34	0.98	fails 2
logistic regression	constructed	1–20	1.33	0.99	fails 2
frozen nonlinear decoder	constructed	1–20	2.13	0.92	fails 4
perceptron in a k -subspace	constructed	1–20	2.06	0.94	fails 4
image data, parameter subspace	measured	1–8	0.87	1.00	complete
image data, parameter subspace	measured	1–20	1.62	0.97	fails 3
image data, function subspace	measured	1–8	0.50	0.99	fails 2, 3, 4

Two rows of table 5 predate the estimator of appendix A. The oscillating matrix was scored with an earlier implementation that applies no Theiler exclusion and clamps the returned value to E , at $E = 25$ and $m = 30$ rather than at the frozen configuration; its grid swept the drive period alongside the estimator parameters, which is the breach of requirement 2 that also disqualifies the two regression systems, and it reports no companion statistics, which is requirement 5. The function-subspace row shares the clamping and the mixed grid, computes one window per run rather than a median over sliding windows, and has no zero-learning-rate control, so requirement 4 is untested there rather than passed.

Both rows are kept because they locate the ceiling, and because the oscillating matrix supplies the synchronisation control of section 5.1, which no other system provides. Neither should be read as a recovery result.

The gap between the two implementations is in their defaults and not in their arithmetic: at matched settings they agree to 0.0 on all thirty cells of a validation set and on twelve further series, both calling the same neighbour kernel. Of the two defaults that differ, the clamp never fired — the largest estimate anywhere in these files is 22.34 against a threshold of 62 — so it is a latent hazard rather than a correction that was applied. The missing exclusion does cost something, but the cost is small and not consistently signed: switching it on moves the held-out error of table 5 by about three hundredths, upward on one seed and downward on the other, and costs the first its perfect rank correlation.

All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld seeds of section 5.4, in the deterministic recurrent regime, and are scored against the measured effective rank rather than against r . Where an aggregation over seeds is needed, the median over the four seeds at each rank is taken before scoring; the choice moves the numbers, so it is stated. Three

figures in this paper describe the same eight-direction measurement under three aggregations, and table 6 separates them; they should not be read as three results.

Table 6: Why three different errors are quoted for one measurement. Each row is the same eight-direction recovery of section 5.4 — withheld ranks, withheld seeds, deterministic recurrent regime — summarised in a different order, and the order is stated wherever one of these figures appears.

aggregation	reported in	error
mean over (rank, seed, observer) runs	section 5.4, table 5	0.87
median over observers of each observer’s error	table 7	0.47
median over seeds and observers at each rank, then scored	table 9	0.457

Table 7: Which scalar observer to prefer, and whether its accuracy could have come from smoothness alone. One row is one observer, except that the two parameter norms share a row; MAE and slope are over the four withheld ranks $r \in \{1, 3, 5, 8\}$ after taking the median over the four withheld seeds at each rank, scored against the measured effective rank. The rank correlation over those four ranks is 1.00 for every observer, so it has no column. The last column identifies the observers whose roughness statistic orders the four withheld ranks perfectly: with $n = 4$ a rank correlation takes only eleven values and nothing short of ± 1 is distinguishable from no ordering, so the column records yes or no and not a number.

observer	family	MAE	slope	roughness orders rank
probe loss	loss	0.30	0.90	no
function-space norm	norm	0.36	0.93	no
margin	function	0.41	0.93	no
parameter norms (two)	norm	0.45	0.85	yes
fixed parameter projection	projection	0.49	0.97	no
gradient projection	gradient	0.56	0.89	no
function-space projection	function	0.70	0.70	no
gradient norm	gradient	1.97	1.65	no
full-batch loss	loss	2.01	1.62	no
instantaneous loss	loss	0.42	0.86	no
probe accuracy	function	degenerate in every window		

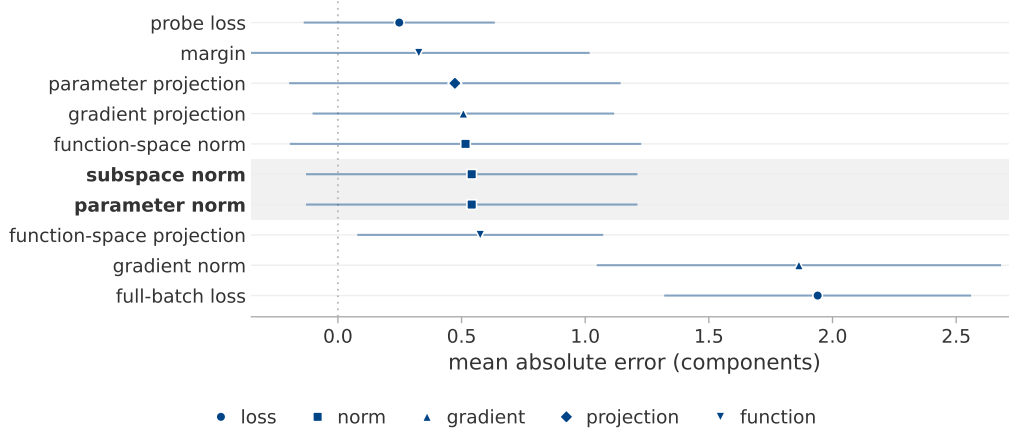


Figure 3: Which scalar log to keep. Ten of the twelve observers of table 2, scored on the fast recurrent case of section 5.4. The point is the mean absolute error of the estimate against the measured effective rank over the seven simulated ranks, taking the median over the four seeds at each; the bar is the mean across-seed standard deviation of the estimate, drawn either side of the point. It is the spread of the estimate and not of the error, so it may extend below zero, and where it does the observer’s error is inside its own seed-to-seed noise. Marker shape is the observer family; the shaded rows are the two parameter norms, the observers a training run ordinarily logs. Eight of the ten lie between 0.25 and 0.58 components with every bar overlapping every other, so this experiment does not choose among them; the gradient norm and the full-batch loss are more than three times worse than the next observer and should not be used. The ordering differs slightly from table 7, which scores only the four withheld ranks and reports the two parameter norms as one row.

Two observers are excluded from every aggregate, for different reasons. Probe accuracy is quantised, which drives the delay vectors together and raises the degeneracy indicator in 100 % of its windows; it is the sole source of the degenerate fraction reported anywhere in this study, every other observer being at zero. The instantaneous mini-batch loss is excluded because it fails requirement 4: at zero learning rate, where the parameters do not move at all, its estimate still climbs from 0.91 to 6.93 across the seven simulated ranks, a range of six components produced entirely by the drive. The other eleven observers are exactly constant under that control, so no estimate is defined for them.

The roughness column makes the main text’s caveat precise. For eight of the nine scored rows the roughness statistic does not order the four withheld ranks, so their agreement with the truth is not available from smoothness. For the two parameter norms it orders them perfectly, and those are exactly the observers a training run ordinarily logs: their result should be read as consistent with, rather than as evidence against, a smoothness explanation.

Table 8: How far recovery extends when twenty directions are available rather than ten. Image data, median over three seeds and eight observers; the truth is the measured effective rank, and the last row is the spread of MG across seeds at fixed rank.

true r	1	2	4	6	8	10	12	14	17	20
MG	0.89	2.35	4.41	6.66	7.38	8.99	10.61	11.40	13.50	15.09
PR _{delay}	2.00	2.29	4.54	6.81	7.95	9.20	10.79	12.15	11.71	13.51
spectral PR	1.01	1.25	2.58	3.68	4.18	5.48	5.98	6.53	8.11	8.38
spread across seeds	0.00	0.20	1.49	2.15	2.11	2.95	1.50	2.84	2.76	3.33

Table 9: Whether any simpler statistic recovers the active dimension as well as the estimator does. All are computed on the image-data system of section 5.4, over its withheld ranks and seeds and under the same convention, at both ranges. Statistics marked † need a neighbour search; the rest are one-line computations on the series or its spectrum. The eight-direction correlations are exact at two decimals — with four ranks, Spearman takes only the values shown — while the twenty-direction ones are given to three, so that 0.998 is not printed as 1.00 beside a correlation that really is one. `paper_tables.py` regenerates every cell.

statistic	eight directions		twenty directions	
	MAE	ρ	MAE	ρ
MG †	0.457	1.00	1.62	0.973
LB †	0.43	1.00	1.36	0.983
TwoNN †	1.81	1.00	3.48	0.889
PR _{delay}	0.89	1.00	2.18	0.979
spectral PR, 256 bands	1.60	1.00	5.29	0.998
spectral PR, 1024 bands	1.35	1.00	4.47	0.998
spectral PR, native resolution	1.03	0.80	2.78	0.934
roughness	3.65	0.20	9.90	−0.568

Table 9 should be read with three things in mind, and none of them favours MG.

First, MG is not the most accurate statistic at either range: the uncorrected per-point average LB is ahead of it at both. The correction of section 3.2 removes a positive bias, and on a saturating estimate that bias was partly cancelling the saturation, so removing it costs accuracy exactly where the estimate is already saturating.

Second, MG is not the best-ordering statistic at twenty directions. The spectral participation ratio at 256 and 1024 bands orders the ranks more faithfully than MG does, at three times the error. If only the ordering is wanted, a statistic requiring no embedding and no neighbour search will supply it.

Third, MG’s advantage over the linear delay participation ratio is a factor of about two at eight directions, not the factor of six that the selection split of appendix C reports. That larger figure is computed on the ranks and seeds the configuration was chosen on and is a property of the selection, not a recovery result.

Dependence on the observer. Running the same torus through two scalar observers that differ only in how evenly they weight the modes settles which statistic is preferable, and the answer is that neither is, in general (table 10). Under an observer that weights all modes equally the spectral participation ratio is the more accurate; under a generic observer with random weights — the form a random projection of a network’s parameters actually takes — MG is. Both orderings are perfect in three of those four cells, so the statistics are separated by absolute recovery and not by rank agreement.

The size of each advantage also depends on the delay lag, which a controlled experiment can choose and a training log cannot. Held at a lag common to both observers rather than at the one best for each, the direction of each advantage survives and its size does not.

Table 10: Whether MG or the spectral participation ratio recovers the active dimension more accurately, and how far the answer depends on the observer and on the delay lag. The two statistics compared are MG and the spectral participation ratio at 256 bands, on one torus; each figure is the ratio of their errors, in favour of the one named in the second column. The first column of figures uses the lag best for that observer, which a training log cannot choose, and the last two hold both observers at a common lag.

observer	more accurate	at its best lag	$\tau = 4$	$\tau = 1$
equal mode weights	spectral PR	3.4	6.6	3.4
random mode weights	MG	4.7	4.7	1.9

E CONDITIONS OF VALIDITY IN DETAIL

This appendix gives the three experiments section 6 summarises — the delay-lag experiment, the nuisance controls and the anisotropy grid — and one it does not: a test that a change in the active dimension is detected when one occurs.

The delay lag. The delay window spans $(E_{\max} - 1)\tau$ samples, and table 11 varies τ on a fixed torus of period 400 at $E_{\max} = 20$, leaving the system unchanged. Below a tenth of a period every rank returns the same tangent constant near 2.5 and the ordering is gone; between three tenths and a half of a period the estimate follows the rank; above one period it diverges. The failure is a collapse of the ordering at one end and a divergence at the other, not a uniform bias.

Table 11: How far the estimate depends on the delay lag. A fixed torus of period 400 at $E_{\max} = 20$; the span is the delay window in units of the oscillation period. The correct rank is the column heading.

τ	span	$r = 1$	2	3	4	6	8
1	0.05	1.42	2.44	2.49	2.63	2.44	2.60
4	0.19	1.43	2.72	3.31	3.83	3.94	4.29
8	0.38	1.43	2.77	4.18	6.16	6.57	8.53
16	0.76	1.44	3.00	5.92	13.66	12.83	20.76
32	1.52	1.46	3.06	9.90	33.28	31.02	63.35

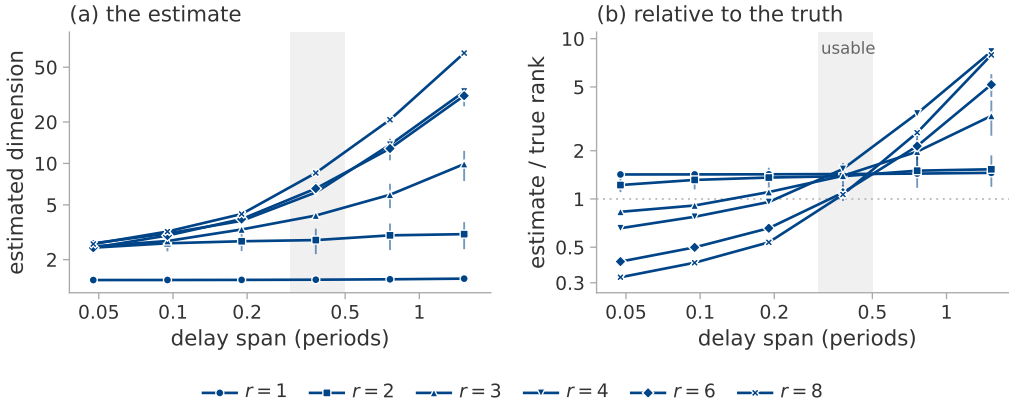


Figure 4: The delay lag, varied on one system: a fixed torus of period 400 samples at $E_{\max} = 20$, two seeds, six fixed lags $\tau \in \{1, 2, 4, 8, 16, 32\}$ spanning 0.05 to 1.52 periods. Marker shape is the true rank, the marker the median of the two seeds and the bar their range. (a) The estimate. (b) The same divided by the true rank, so one reference line replaces six; the shaded column is three tenths to a half of a period, where the estimate is within one component at $r = 6$ and $r = 8$ and overshoots by one to two at $r = 3$ and $r = 4$. Across this range the estimate at $r = 8$ moves by 60.7 components and at $r = 1$ by 0.03. Table 11 gives the same numbers. A seventh setting, in which τ is taken from the autocorrelation, is omitted because its span differs between the two seeds.

Nuisance factors. Table 12 varies seven factors that leave the active dimension at four. Rotating the coordinates is scored on the fixed parameter projection alone, since the other eleven observers are invariant to it by construction, so that row measures a property of the observer rather than of the estimator.

Table 12: Shift in the estimate produced by factors that leave the active dimension unchanged, median over five seeds and twelve observers. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale against which the others should be read. The last column is a percentage of the 3.95-component fall that the change-detection experiment below observes for a true four-component change. A constant rescaling of the observer, which the scorer standardises away, moves the estimate by 0.000 on every observer and is omitted.

factor	$ \Delta MG $	as % of a real four-component change
none (baseline)	0.21	5
learning rate halved	0.16	4
drive band up one octave	0.49	12
drive band down one octave	0.50	13
coordinates rotated	0.86	22
observer gain ramped $10\times$	1.16	29
state amplitude ramped $4\times$	1.32	33

Detection of a change. Recovery of a static level does not establish that a change is detected. We run three consecutive segments $r_{\text{high}} \rightarrow r_{\text{low}} \rightarrow r_{\text{high}}$ with nothing else altered, re-measuring the ground truth on each. In the deterministic recurrent regime a true fall of 3.99 components is observed as 3.95, is detected by every usable observer in both directions, and recovers to within 0.01 of the level before it. Under stochastic excitation the same true change is observed as 0.15.

The rule fires when the estimate falls below the median of the pre-switch segment by three times that segment’s window-to-window scatter, for two consecutive windows. The threshold uses the pre-switch segment alone, never the midpoint of the two observed levels, which would give the detector the answer and would decide arbitrarily the case in which the two levels coincide.

Table 13 gives the three schedules on the parameter norm. The level error is 0.89 at the high level and 0.22 at the low one: the estimate is more accurate the fewer directions are active.

Table 13: Whether a change of the active dimension is recovered at both levels and after it reverses. Three schedules $r_{\text{high}} \rightarrow r_{\text{low}} \rightarrow r_{\text{high}}$, on the parameter norm. At $8 \rightarrow 3 \rightarrow 8$ the high level is under-reported by a component, which is the saturation of table 8 seen from another direction.

schedule	high	low	high again
$4 \rightarrow 1 \rightarrow 4$	4.77	1.22	4.78
$6 \rightarrow 2 \rightarrow 6$	6.90	2.12	6.91
$8 \rightarrow 3 \rightarrow 8$	6.97	3.64	6.97

The response time cannot be quoted from this experiment. With a window of 8 000 samples and a stride of 2 000 the two observed lags, 999 and 1 999 steps, are the two floors of the measurement grid rather than measured values. The experiment does establish that the estimate crosses the threshold on a window still containing 87.5 % pre-switch data.

Anisotropy. Table 14 gives the measurement of section 6.4 in full: the system of section 5.1 with mode l driven at amplitude q^l , three seeds, observed through the squared Frobenius norm and estimated with the frozen eight-direction configuration. The manifold dimension is r in every row.

Table 14: Whether the estimate follows the number of modes or the effective rank when the two disagree. An r -torus with geometrically decaying amplitudes; r is the manifold dimension and is constant along each block, q the amplitude decay factor; PR is the participation ratio of the covariance of the r oscillating coordinates. Median over three seeds.

r	q	PR	MG	PR _{delay}
2	1.0	2.00	2.30	3.12
2	0.7	1.79	2.33	3.28
2	0.5	1.47	2.35	3.18
4	1.0	4.00	3.82	6.44
4	0.7	2.60	3.89	4.85
4	0.5	1.65	4.06	3.11
6	1.0	6.00	8.76	8.23
6	0.7	2.84	6.90	5.53
6	0.5	1.67	4.83	3.21
8	1.0	8.00	8.88	8.76
8	0.7	2.90	5.26	5.19
8	0.5	1.67	4.86	3.00

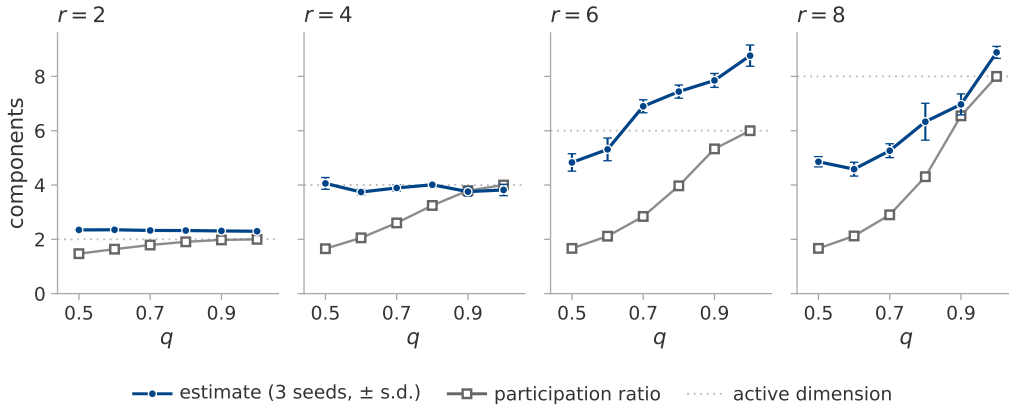


Figure 5: Anisotropy separates the two candidate estimands, and the estimate follows the number of modes. The system of section 5.1 with mode l driven at amplitude q^l , three seeds, observed through the squared Frobenius norm and estimated with the frozen eight-direction configuration; one panel per r . The manifold dimension is r in every panel, drawn dotted: the orbit closes onto an r -torus whatever q does. Filled circles are the estimate, the median over seeds, with their standard deviation as bars; open squares are the measured participation ratio of the covariance, which agrees with the closed form $(\sum_l a_l^2)^2 / \sum_l a_l^4$ to two decimals in all twenty-four cells and so carries no spread — the check that the construction does what it claims. Where the estimate stays flat while the participation ratio declines, the estimator is reporting how many coordinates the set has rather than how evenly they are used. Table 14 gives twelve of these points in digits.

At two and four directions the estimate is flat in q while the participation ratio declines, as intended. At six and eight it declines too, and the reason is visible in the amplitudes: at $q = 0.5$ and $r = 8$ the weakest mode is driven at 0.008 of the strongest, far below the neighbourhood scale the estimator works at, so the torus it sees really does have fewer dimensions than the one that was built. The estimator therefore reports the number of modes it can resolve, which is neither the manifold dimension of the construction nor the participation ratio of its covariance, and coincides with both only when the amplitudes are equal.

F SYSTEM CONSTRUCTIONS

Three constructions are described here: the image-data system of section 5.4, the function-space variant scored beside it, and the same system at twenty available directions.

The image-data system of section 5.4. The data is partitioned into twelve fixed groups, the same partition at every rank so that the partition is not itself a function of r , and the loss weights of r of them are modulated by r rationally independent sinusoids filling one octave. Modulating group j tilts the gradient along a direction ϕ_j . These directions are neither orthogonal nor of equal effect, since random data groups have correlated gradients, so a mixing matrix $\text{pinv}(\Phi)Q$ with Q an orthonormal basis of $\text{range}(\Phi)$ orthogonalises them, and the per-mode scalar response gain of the linearised update is divided out. Without this correction the trajectory participation ratio falls well below r .

Table 15 verifies that the construction achieves what it intends, and two of its rows are results rather than settings. A decaying transient does not recur whatever r is, since a converging trajectory is a curve; and plain mini-batch descent remains at 6.2 regardless of r , because its noise rank is fixed by the per-example gradient covariance of the data and is not a free parameter of the experiment. Projecting real mini-batch noise onto r directions recovers only part of the intended rank, which is why that regime is scored against its measured value and not against r .

Table 15: Whether each construction excites the rank it claims. Measured trajectory participation ratio by excitation regime and nominal rank, median over the four seeds, for the ten available directions of section 5.4. Every estimate in this paper is scored against these values, never against r .

regime	$r = 1$	2	4	6	8	what is excited
qp	1.00	2.00	4.00	6.00	8.00	r sinusoids, fast drive
qp_slow	1.00	2.00	4.00	6.00	7.99	the same at a training-log timescale
qp_nopre	1.00	1.99	3.99	5.89	7.55	qp without the preconditioner
mixed	1.00	2.00	4.00	6.00	7.99	qp plus noise in the same directions
noise	1.00	2.00	3.99	5.99	7.98	injected rank- r gradient noise
noise_nopre	1.10	2.08	3.74	5.47	6.43	the same without the preconditioner
batch_proj	1.00	1.79	3.25	4.53	5.61	real mini-batch noise projected to rank r
batch	6.20	6.25	6.21	6.26	6.20	plain mini-batch descent
gd	1.05	1.05	1.05	1.03	1.06	a decaying transient
qp, $\eta = 0$	1.00	1.00	1.00	1.00	1.00	requirement 4 control

In the last row the learning rate is multiplied by zero, so the active dimension is identically one at every nominal rank and an observer whose estimate still tracks r there is reading the drive rather than the optimiser.

Two properties of the backbone bear on the construction. The functional rank is 10 at every rank, seed and regime, with a participation ratio between 8.40 and 9.03; it belongs to the frozen backbone and does not depend on the excitation. And the condition number of the drive-direction matrix is 1.00 at $r = 1$ by construction and has a median of 13.3 at $r = 8$, which is the anisotropy the mixing matrix exists to remove.

The function-space construction. A local adapter is built around a trained network of widths 64, 16 and 10, and its function-space Jacobian is QR-whitened, so that its first r columns have rank r at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it by exact backpropagation through both layers. The functional, trajectory-covariance and update-covariance ranks were all verified to equal r .

The twenty-direction system. The same construction at twenty available directions reproduces the nominal rank to within 0.008 components over $r = 1, \dots, 20$ in the recurrent regime, and to three decimals up to $r = 10$. The other regimes behave as at ten directions: injected noise tracks r closely, projected mini-batch noise reaches only 9.4 at $r = 20$, and the decaying transient stays between 1.02 and 1.09 throughout.

G DIAGNOSTICS AND FIGURES FOR THE GROKING APPLICATION

This appendix carries the diagnostic values behind section 7.1, a map of every training log in the plane of those diagnostics, and the label-matched pairs of section 7.1 beside the loss curves that account for them.

Table 16: Whether either grokking setting is admissible. The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Every cell is the median over the run’s sliding windows. Seven of the seventeen runs of table 24 are shown; figure 6 plots all seventeen, and the ranges quoted in section 7.1 are over all five regularised runs, not the three shown here. Runs that have not generalised within the budget are censored, not negative.

run	generalises	MG	ρ_{ident}	roughness	PR_{delay}	crossings
transformer, positive control	yes	15.31	1.29	0.030	1.02	56
transformer, reduced data, s0	yes	4.93	1.60	0.068	1.52	146
transformer, reduced data, s2	no	4.47	1.51	0.149	3.94	151
transformer, no weight decay	no	28.08	1.00	< 0.001	1.00	2
perceptron, $n + m$	yes	19.54	1.00	0.001	1.00	2
perceptron, $n \cdot m$	yes	19.63	1.00	0.001	1.00	2
perceptron, $n^3 + nm^2 + m$	no	22.43	1.00	< 0.001	1.00	2

Figure 6 places every training log of section 7 in the plane of the two diagnostics, against the region in which every accurate case of section 6 fell. No run of either setting is in it: the regularised mini-batch runs are unstable in the embedding dimension, and the full-batch and zero-weight-decay runs are stable only because their series is monotone.



Figure 6: Whether any training log of either setting is admissible. Every log of section 7 in the plane of the two diagnostics, one point per run: seven transformer runs of 120 000 steps observed through the parameter norm, and ten perceptron runs of 100 000 steps observed through the training loss. Colour is the regime of table 1 and marker shape the setting: the five regularised mini-batch runs are stochastic, the ten full-batch perceptron runs and the two zero-weight-decay transformer runs transient. The vertical axis is the identifiability ratio, near unity when the estimate does not depend on the embedding dimension; the horizontal axis is the trend-crossing count, which tests non-monotonicity of the series and not recurrence in the reconstructed space. The grey lines are the edges of the shaded rectangle: to their left too few crossings, above them instability in E . Twelve of the seventeen runs coincide at exactly two crossings with ρ_{ident} between 0.9975 and 0.9992; those two groups are drawn with a small offset either side of their true abscissa, which the grey tick marks, and labelled with their multiplicity. The shaded rectangle is where every accurate case of section 6 fell, $\rho_{\text{ident}} \leq 1.10$ and more than eight crossings. It describes the calibration and is not a validated decision rule. The two settings fail in different corners of it without reference to any label.

Figure 7 shows the separation of section 7.1 and the mechanism behind it: the two members of a pair differ in the shape of their training curves long before any question of dimension arises.

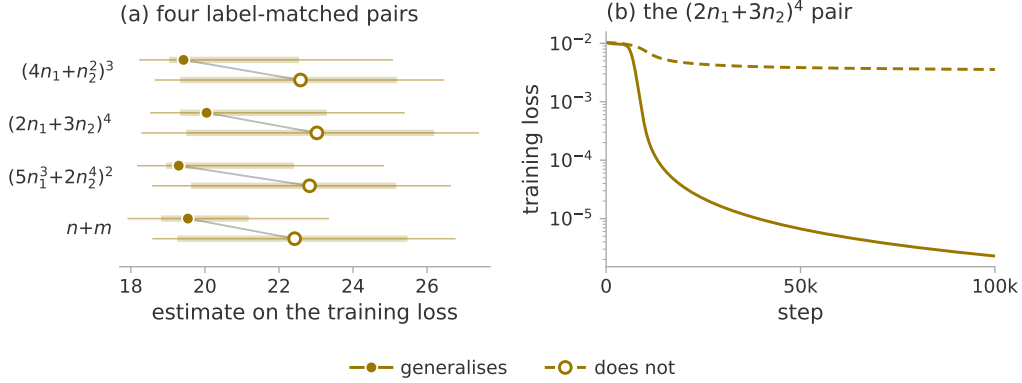


Figure 7: Whether the estimate separates label-matched pairs, and what accounts for the separation. (a) The estimate on the training loss for the four pairs of section 7.1: three modular polynomials at $p = 97$, each against itself perturbed by one monomial, and modular addition against $n^3 + nm^2 + m$. Both members of a pair reach full training accuracy under identical optimisation on identical data budgets. All eight runs are full-batch perceptrons, which figure 6 places in the transient regime, so the figure carries that one hue and generalisation is carried by fill and dash. The marker is the median over the seven sliding windows, the thick bar their interquartile range and the thin line their full range. The medians separate the two members of every pair, by 2.88 to 3.54 units and with no overlap between the two groups of medians, but the window-level ranges overlap heavily, because the estimate drifts upward over training in every run rather than scattering about a level. (b) The $(2n_1+3n_2)^4$ pair, the second row of (a). The representable target drives the training loss to 2.3×10^{-6} while the perturbed one stops at 3.6×10^{-3} , three orders of magnitude short despite fitting the training set completely, so the two curves differ in shape rather than in any attractor dimension. Across the four pairs that gap runs from 2.7 to 3.2 orders.

H THE EFFECTIVE-RANK COLLAPSE, RUN BY RUN

Table 17 states the measurement of section 7.2 for every run. The plateau is the median of the 4 000 steps before the generalisation step, the dip the minimum within $\pm 4\,000$ of it, and “at” the position of that minimum relative to it; the controls, which have no generalisation step, are measured in the window of their matched weight-decay run.

The last column is the largest value the statistic reaches after its dip, which is what “re-expands” means: in parameter space it exceeds the plateau by an order of magnitude in every run, in function space it returns to about the plateau and exceeds it in three of the four.

Table 17: Plateau, dip and recovery per run, in function space and in parameter space. Depth is plateau divided by dip. The two control rows in each block are measured in the aligned window; their deepest fall anywhere in their own budget is 3.15 and 7.06 in function space and 1.66 and 2.37 in parameter space, which is the comparison section 7.2 also reports.

space	run	plateau	dip	depth	at	post-dip max
function	mod_wd1	23.12	4.42	5.23	+595	24.95
	mod_wd1_s43	33.06	2.71	12.21	+235	27.82
	mod_wd1_s44	28.26	3.65	7.74	+515	31.58
	s5_wd1	21.15	3.41	6.20	+712	24.21
	mod_wd0	5.79	3.60	1.61	—	—
	s5_wd0	11.75	8.86	1.33	—	—
parameter	mod_wd1	3.09	1.09	2.84	+1 095	35.34
	mod_wd1_s43	2.70	1.15	2.35	+935	38.04
	mod_wd1_s44	2.71	1.13	2.39	+1 315	37.34
	s5_wd1	7.00	1.60	4.38	+862	13.43
	mod_wd0	30.10	27.70	1.09	—	—
	s5_wd0	5.59	4.53	1.24	—	—

I THE TRAJECTORY SKETCH

The direct measurement of section 7.2 needs the trajectory itself, which is the whole parameter vector at each of two to three thousand logged steps. Every logged step is therefore compressed by a CountSketch (Charikar et al., 2002; Weinberger et al., 2009) to $\mathbb{R}^{1024}$, with two independent hash families. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length. The normalisation is necessary: the raw logit scale is mechanically coupled to weight decay, so a signal read off it could not be distinguished from a detector of weight decay.

We do not verify a spectral guarantee for this. The elementary bound for a sketch of this kind is on pairwise inner products (Weinberger et al., 2009), and a participation ratio is a functional of the whole spectrum, on which an entrywise Gram bound is weak. A subspace-embedding guarantee does exist for CountSketch (Clarkson & Woodruff, 2013) and would apply here, since a sixty-sample window has rank at most sixty, but it requires a rank we do not measure per window. The two hash families supply a direct estimate of the error instead: every reported value is the mean of the two, and their disagreement is recorded with it. Over the published transformer windows that disagreement is 1.1 % of the value at the median and 8.4 % at its worst, which is the number the reader should use, and it is smaller than every effect section 7.2 reads.

Three checks stand behind the numbers. *The sketch is accurate on synthetic trajectories of known rank.* Table 18 pushes such trajectories through it and compares against the same statistic computed on the uncompressed trajectory: the sketch is accurate to 0.04 components throughout, and the shortfall against the nominal rank at 5 and 10 is present without any compression, so it belongs to the participation ratio on a sixty-sample cloud and not to the sketch. Nothing above rank ten is tested, however, and the plateaus in figure 2 are higher than that.

It agrees with exact arithmetic on a real run. On the perceptron architecture, where the parameter count is small enough to allow it, the participation ratio computed on all 145 500 raw parameters agrees with the sketched value to three decimals.

The observer does not perturb training. The log is required to be bit-identical with and without it, over four hundred steps at reduced probe size. That check matters because the task module seeds one global random stream that the train-validation split, the initialisation and the mini-batch order all continue: a single stray draw changes the initial weights and can destroy grokking.

Table 18: What the compression costs. The sketch against the same statistic computed without it, on synthetic trajectories of 60 samples and known rank in 145 500 dimensions. The last column gives the cost of the compression; the gap between the first two gives the cost of a sixty-sample participation ratio, which is not attributable to the sketch.

true rank	PR, uncompressed	PR, sketched	sketch – uncompressed
1	1.00	1.00	0.00
2	1.99	1.98	−0.01
5	4.42	4.46	0.04
10	8.61	8.63	0.02

Windows are sixty logged rows, which is 600 optimiser steps for the modular runs and 300 for S_5 , and are labelled by their centre, so a feature at a given step is located to within half a window and the window is not causal. Five statistics are computed per window in each space, of which the paper reports two: PR^{det} , the participation ratio of the positions after removing a per-window linear trend, since a steady drift is rank one and would otherwise mask everything else; and PR^{upd} on the increments block-averaged over five logged steps, to suppress the mini-batch noise floor. Beside them the total displacement is recorded, as the null for a rank that falls simply because the trajectory stopped moving.

J THE FULL-BATCH SETTING

Repeating the direct measurement on the perceptron setting gives a result that must be read as a limit of the measurement rather than as a negative. Under full-batch gradient descent every window of every run has a participation ratio between 1.000 and 1.362 on the four statistics the comparison uses, and between 1.000 and 2.171 over all eight recorded columns. This is not an artefact of the sketch, which appendix I checks against exact arithmetic on the same architecture.

A deterministic update makes the trajectory a smooth curve, and over a 600-step window a smooth curve is straight to within the measurement precision, so what the statistic reports is the curvature of the trajectory at the scale the window sets, and at that scale there is almost none. The statistic has no range in which a collapse could appear, so the absence of a feature is not evidence of its absence: the window must be set in units of the curvature scale of the trajectory.

Repeating the measurement against window length confirms the diagnosis and withdraws the opening it appeared to leave. Two runs of 150 000 steps, a generalising one and its label-matched control, are measured over windows spanning 600 to 120 000 optimiser steps, with the window lengthened by spreading sixty samples further apart rather than by adding samples, so that the bound on the statistic and its noise floor stay exactly where the published measurement put them and only the span changes. Table 19 gives the result. The statistic does leave the floor, at a scale of about 10^4 steps, two orders of magnitude above the window that produced the 1.000–1.362: the absence of a feature there was a resolution limit and not a fact about the run.

It is not, however, a hidden signal. At 12 000 steps one localised maximum appears, and both runs have one of the same size (table 19); function space agrees, with more range and no more separation, 2.618 against 2.145. Nor can the maximum be placed relative to either milestone: the two centres are adjacent points of a 3 450-step grid and their windows overlap over 6 900 to 15 250 steps. Measured against memorisation the two offsets are +3 600 and −450, a difference this design cannot resolve — a 12 000-step window cannot place a feature to better than the interval between memorisation and generalisation.

The claim the experiment does support is the negative one: at the window where the statistic has range, the generalising run and its label-matched control are not separated by it. Beyond 30 000 steps the ordering reverses and the control moves further than the generalising run, because the window is then a large fraction of a run whose weight norm is still climbing and the statistic is reading that drift. The full-batch trajectory therefore carries no rank signature of generalisation at any window length between 600 and 120 000 steps: the collapse of section 7.2 belongs to the regularised stochastic setting, and a trajectory participation ratio must be controlled for window length as well as for batch size before anything is read from it.

Table 19: Whether the full-batch statistic has any range at longer windows. Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window. Sixty samples per window at every length. The 600-step row is the measurement reported above, on the same configuration and seed.

window (steps)	generalises		control, never generalises	
	median	max	median	max
600	1.002	1.035	1.001	1.011
3 000	1.070	1.253	1.003	1.092
6 000	1.028	1.537	1.005	1.360
12 000	1.011	1.877	1.020	1.843
30 000	1.007	2.111	1.113	1.725
60 000	1.021	1.660	1.335	1.975
120 000	1.256	1.659	1.929	2.337

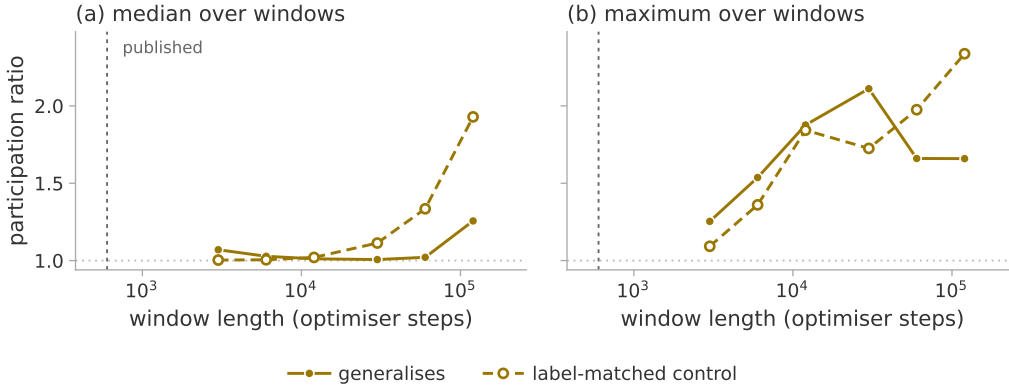


Figure 8: The trajectory participation ratio against window length, on the full-batch pair of this appendix: modular addition, which memorises at 9 200 and generalises at 11 950, and its label-matched control, which memorises at 9 800 and does not generalise. The statistic is $PR^{\det}$ on the sketched parameter trajectory over windows of 3 000 to 120 000 optimiser steps, the window lengthened by spreading sixty samples further apart rather than by adding samples, so that the bound and the noise floor stay where the published measurement put them and only the span changes. (a) The median over a run’s windows, (b) the maximum; the dotted line at 1 is the floor a straight trajectory returns, and the dashed vertical the 600-step window of section 7.1, which is not part of this series. Both runs are transient, so the figure carries one hue and the control is drawn open and dashed.

Introducing mini-batches confirms the diagnosis. The same statistic then ranges over more than a decade while the generalisation step moves by one logged step, so the learning is unchanged and the statistic has moved by an order of magnitude. Any claim read from a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

K DECLINES OF THE ESTIMATE THAT ARE NOT GENERALISATION

A decline of the estimate is not specific to generalisation. In a 200 000-step run at $p = 211$ with no weight decay, which memorises at step 1 800 and ends at 3.6 % validation accuracy, a decline is detected 49 250 steps after memorisation. The amplitude control of section 6.3 supplies a mechanism that needs no dimension at all: a ramped observer gain moves the estimate by 1.16 components, and this run’s parameter norm grows from 43 to 126 over the budget. It is the growth section 7.2 reports in the two weight-decay-zero controls, acting through the observer.

That run does not show a simplified trajectory. Its diagnostics place it in the transient regime (table 16), where the estimate is not a dimension, so a decline in it is a decline of a quantity we have argued

should not be read as one. The two must be kept apart: the direct measurement of section 7.2 is an effective rank of the trajectory itself, and it declines in four generalising runs and not in their controls, whereas the estimate on a log, at the frozen window and in these settings, is neither a dimension nor an effective rank and declines in at least one run that does not generalise. The defensible claim is therefore the weaker one: a decline on a log is not evidence of generalisation.

Section 7.3 does not overturn it. There the log estimate declines where the trajectory does, in four runs and in neither control, but at a window two orders of magnitude shorter than this one and under a configuration chosen for that window; the level is no more readable there than here, and what is read is a change against the run’s own distribution of changes. Nor is the absence of generalisation within a budget evidence that a decline was a false alarm: two reduced-data configurations against which earlier specificity figures were computed generalise once the budget is extended, so such runs are censored and no specificity figure may be computed against them.

L RESOLUTION OF A WINDOWED ESTIMATE ON A TRAINING LOG

The claim of section 7.1 that a change within a run could be read against a fixed configuration is testable on the seven extended runs, and the test returns a resolution limit rather than an answer. The frozen configuration takes a window of a third of each record: 4 000 logged samples, which at the logging stride is 39 990 optimiser steps, advanced by 10 000 steps. That is nine windows per run and sixty-three weight-norm windows in all, and it localises a feature only to $\pm 19\,995$ steps, sixty-seven times coarser than the direct measurement of section 7.2.

The consequence is that the alignment figure 2 performs cannot be performed here. Of the twenty-seven window centres belonging to the three generalising runs, one lies within 5 000 steps of its own generalisation step and two within 10 000; no two runs share an offset, so no offset has more than one run defined and no pooled mean exists. For the positive control the entire pre-generalisation phase is thirteen per cent of one window; for the reduced-data run that generalises at 110 940, no window centre falls after the transition at all.

The runs are consistent with that. The one run with a centre near its generalisation step falls by 3.54 components over the next stride, three times the floor — but the step lies inside four of its nine windows, and the falling windows are those in which its parameter norm is itself moving fastest, which is the ramped gain of section 6.3. The other two runs show nothing.

Nor is any of this specific. Across the fifty-six stride-to-stride changes, six of twenty-four clear the floor in the runs that generalise and six of thirty-two in the runs that do not. The largest change in the set, 6.36 components, is in a run that does generalise, but it falls nearly ninety thousand steps after its generalisation step; the next largest, 5.20, is in a run that ends at seven per cent validation accuracy. The tightest comparison available is three runs of one configuration, of which one generalises: their ranges are 0.77, 0.78 and 0.67 components, and nothing separates them.

This is the argument of appendix J in a second place. At the window the configuration provides, the statistic has no range in which a feature localised to the transition could appear, so its absence is not evidence of absence. Testing the claim needs a window set in units of the width of the transition, which for these runs is hundreds of steps and not tens of thousands: a re-run of the estimator, not a re-reading of what we have.

The matched-window re-run. Section 7.3 performs it, on the six runs of figure 2 rather than the seven of this appendix, because those are the runs whose trajectory was stored and only a paired comparison answers the question. The window is 60 samples throughout, which is 600 optimiser steps for the modular runs at their stride-10 logging and 300 for the S_5 runs at stride 5, centred on the midpoints of the windows of section 7.2; since those are 600 and 295 steps wide, every log window matches the one it is paired with to within two per cent. Four things bear on how the result should be read.

The configuration is not the frozen one, and cannot be. At 60 samples the frozen delay span, $(20 - 1) \times 4 = 76$, exceeds the window. The grid crosses $E_{\max} \in \{4, 6, 10\}$, $\tau \in \{1, 2, 4\}$, $k \in \{5, 20\}$ and both Theiler rules, thirty-six cells, of which twenty-six are long enough to return a value. Twenty-five of those separate the four generalising runs from the two controls; the exception is $E_{\max} = 6$, $\tau = 1$, $k = 5$ under the embedding rule. The headline cell is named by the rule of

section 7.3 and is not the best cell: $E_{\max} = 4$, $\tau = 4$, $k = 20$ gives a shallowest generalising fall of 2.12 against its 1.91.

The admissibility diagnostics do not survive the shortening. The identifiability ratio needs $2E_{\max}$. At $E_{\max} = 10$ the doubled delay span does fit the window, but the Theiler exclusion then leaves two usable points against $k = 20$, so the headline cell has no ratio. Where it can be computed, at $E_{\max} \leq 6$, it reads 0.97 to 1.00 under the frozen Theiler rule and 1.29 to 1.62 under the other, and settles nothing. That is why section 7.3 reads a change and not a level.

Linear detrending removes the effect, and the surrogate control does not. Repeating every cell on the linearly detrended window — the removal PR^{det} performs per coordinate — leaves no cell separating the two groups. The two removals are not comparable: on a 1024-dimensional sketch a per-coordinate linear detrend takes one direction of many, while on a scalar it takes most of the series. The surrogate control keeps the shape and destroys only the fine structure, and it is the one that discriminates. Both are reported in table 20.

The observer matters. The result above is the parameter norm. On the training loss ten of the twenty-six cells separate the groups and the surrogate control passes in the S_5 run only; on the validation loss fifteen do, and that observer defines the event being aligned on, so it is listed for completeness and not read.

The controls have almost no dynamic range. At this window the baseline belongs to the configuration and not to the run: on a standardised 60-sample straight line equation (3) returns 2.7733, and every run's pre-transition level lies within 0.28 of that. The modular control's parameter-norm log is straight to 0.7 % of its own spread across the whole comparison interval, so its estimate cannot move — a span of 0.0029 components over fifty windows, against 2.05 to 2.56 in the generalising runs. Its $D = 1.00$ therefore records that the observer offered the estimator no local structure to respond to, not that a dimension was measured and found unchanged: a statistic with three thousandths of a component of range could not have shown a two-fold fall whatever the trajectory did. The S_5 control is a weaker case of the same thing, 3.2 % residual and a 0.22-component span.

This removes any reading of the two controls' D and quantile columns as informative dimensional negatives. It leaves the surrogate control untouched, since there each run is compared only against surrogates of itself, and it leaves the four generalising runs untouched; and it reinforces the point already made, that the separation between the groups is mechanically a separation in local roughness.

Table 20: Whether the log estimate falls at generalisation, and whether surrogates preserving the observer's shape reproduce that fall. The matched-window re-run of section 7.3, on the parameter-norm log, at the headline cell $E_{\max} = 10$, $\tau = 1$, $k = 20$, autocorrelation Theiler. D is the median estimate over $[t_{\text{gen}} - 3000, t_{\text{gen}} - 1000]$ divided by the minimum over $[t_{\text{gen}} - 1000, t_{\text{gen}} + 2000]$; the quantile is its rank among the same statistic evaluated at every window centre after memorisation in that run; the floor is where the minimum falls. The three p columns are the fraction of 39 surrogates, at each smoothing length in samples, whose D is at least the observed one, as a median over five independent seedings. The spread across seedings is why the median is reported: a p resolved to fortieths from one draw is not a quotable number. Over all five, the largest generalising value is 0.125 and the smallest control value 0.500. Controls are aligned on the generalisation step of their matched run.

run	D	quantile	floor	p_{101}	p_{201}	p_{401}
modular, seed 42	2.36	0.99	+95	0.025	0.025	0.025
modular, seed 43	1.91	0.93	+35	0.100	0.050	0.050
modular, seed 44	2.14	0.92	+315	0.050	0.025	0.025
S_5 composition	2.52	0.89	+712	0.025	0.025	0.025
modular, no decay	1.00	0.46	−905	0.950	0.975	1.000
S_5 , no decay	1.05	0.38	+1112	0.650	0.850	0.975

The same comparison on the nulls the estimator carries: PR_{delay} gives 1.01 to 1.52 on the four generalising runs against 1.00 on both controls, and the roughness gives 6.5 to 55.4 against 1.39 and 1.23. The linear participation ratio therefore responds less than the estimator and the roughness far more, which is the reading section 7.3 gives.

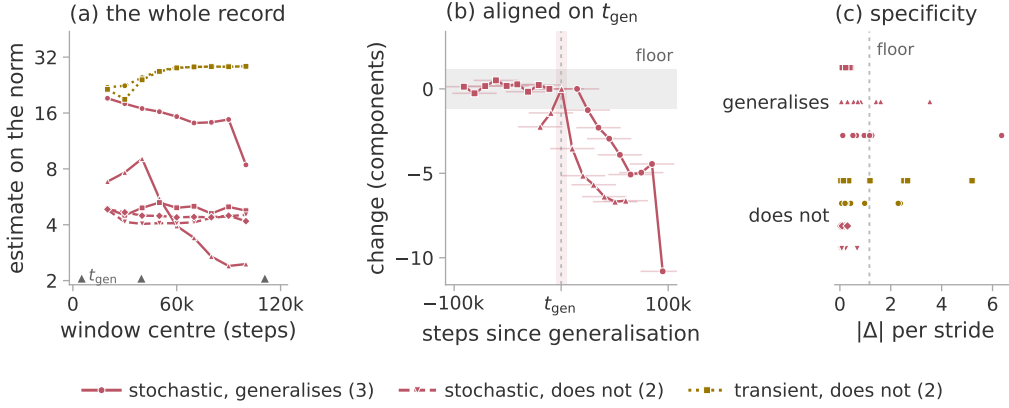


Figure 9: What a frozen-window estimate can and cannot localise. The windowed estimate on the parameter norm for the seven extended 120 000-step runs, under the frozen configuration of section 7.1: nine windows per run of 39 990 optimiser steps each, advanced by 10 000, plotted at their centres. Colour is the regime of table 1, five runs stochastic and two transient, none recurrent. (a) All sixty-three windows, with the three generalisation steps marked on the axis. (b) The three generalising runs aligned on their own generalisation step, each point a change from the window before it and its bar the span it summarises, so consecutive points share three quarters of their span and nothing is localised to better than $\pm 20\,000$ steps. The grey band is the ± 1.16 -component floor of section 6.3 and the shaded column is $\pm 5\,000$ steps of t_{gen} , which one of the twenty-seven centres reaches. (c) Every stride-to-stride change, one sub-row per run so that coincident changes stay countable, against the same floor.

M THE REPRESENTATION DIMENSION IN CLOSED FORM

The perceptron setting of section 7 makes the learned representation analytically available, and three different numbers can be read off it. They are not commensurable and the literature does not always separate them.

The *mode count* is $(p + 1)/2$, or 49 at $p = 97$: the number of distinct Fourier frequencies the generalising solution of Gromov (2023) uses. The *order parameter* is the mean inverse participation ratio of the first layer’s Fourier spectrum, which is 1.000 for that solution because each neuron carries a single frequency, and about 0.041 at random initialisation; it measures how concentrated each neuron is, not how many modes exist. The *effective rank* is the participation ratio of the singular values of the first-layer weight matrix, which for the analytic solution is 148.8: that matrix is 500×194 and full rank, because each non-zero frequency contributes a sine and a cosine direction in each of the two input blocks, so 148.8 says how evenly those 194 directions are used and does not count anything. Only the first is a dimension in the sense the closed form fixes. Two things follow from measuring the other two, and both qualify readings of the representation dimension elsewhere.

The representation is not complete at the transition. Table 21 follows the order parameter through a modular addition run. It does not jump at the transition: it begins rising at memorisation, passes smoothly through, and goes on rising for tens of thousands of steps afterwards without reaching the analytic value. Generalisation therefore arrives well before the representation is complete, and requires it only to be periodic enough.

Table 21: Whether the representation is complete when generalisation occurs. The order parameter through one modular addition run at $p = 97$, full-batch gradient descent with no weight decay, against the analytic solution of Gromov (2023). The run generalises at step 11 950, and the budget ends some 87 000 steps after validation accuracy first reaches 100 %.

point in the run	order parameter
just after memorisation	0.067
just before generalisation	0.189
validation accuracy first reaches 100 %	0.263
end of the budget	0.595
the analytic generalising solution	1.000

The effective rank of the same layer carries almost no signal at all: it reads 139.1 at random initialisation, within seven per cent of the analytic 148.8 before any training, then spans 132.6 to 150.9 over the run without being monotone and overshoots the reference before falling back. Landing near the reference is therefore not evidence of anything.

The order parameter is only interpretable against its own reference. Table 22 gives the measured value beside each task’s own closed form. Two of the tasks that generalise have a reference near the floor, because the representation is periodic in $g_1(n)$ while the spectrum is taken over the raw index n , and g_1 is not linear there. Read against modular addition’s 1.000, a measured 0.044 says nothing was learned; read against that task’s own 0.062, it says the run has essentially converged. The tasks that never generalise remain at the initialisation floor, and the analysis returns no reference for them rather than substituting another task’s.

Table 22: Fourier inverse participation ratio of the first layer at the end of training, against each task’s own closed form. All runs are full-batch gradient descent with no weight decay at $p = 97$. A dash marks a task for which the source papers give no closed form; the four tasks below the rule, whose grokking step reads “none”, are the ones that never generalise.

task	measured	own reference	grokking step
$n + m$	0.595	1.000	11 950
$n - m$	0.582	1.000	11 570
$n^2 + m^2$	0.093	0.052	10 380
$n \cdot m$	0.073	—	11 600
$(4n_1 + n_2^2)^3$	0.243	1.000	12 560
$(2n_1 + 3n_2)^4$	0.283	1.000	7 750
$(5n_1^3 + 2n_2^4)^2$	0.044	0.062	6 430
$(4n_1 + n_2^2)^3 + n_1n_2$	0.040	—	none
$(2n_1 + 3n_2)^4 - n_1^2$	0.043	—	none
$(5n_1^3 + 2n_2^4)^2 - n_2$	0.039	—	none
$n^3 + nm^2 + m$	0.040	—	none

N THE EXCLUSION CAP AT THE TWENTY-DIRECTION CONFIGURATION

The Theiler exclusion is meant to remove neighbours that are close in time rather than in phase space, and the rule this study asks for is $W_T = \max((E - 1)\tau, t_{acf})$: at least the full span of a delay vector, so that two neighbours cannot be near each other merely because their windows overlap. The implementation caps W_T at 150 samples, because the neighbour query cost is linear in it.

At the eight-direction configuration the embedding span is $(20 - 1) \times 4 = 76$, so the cap binds only where the autocorrelation time exceeds 150: never on the calibration systems, whose autocorrelation time is one to three samples, but on five of the seven parameter-norm records of section 7, where it is 161 to 858. Those five were scored at $W_T = 150$ rather than at the value the rule asks for, in the direction that inflates the estimate. At the twenty-direction configuration the span is $(40 - 1) \times 16 = 624$, so the cap binds unconditionally and every row of those scores records

$W_T = 150$, under a quarter of what the rule asks for. This is a departure from the stated protocol, not a choice we made deliberately, and it is why section 5.4 marks that result exploratory.

Table 23 measures what it costs, by re-scoring the stored trajectories at both exclusions. Nothing is regenerated, so the systems cannot drift between the two settings.

Table 23: What the exclusion cap costs. The twenty-direction system re-scored at the capped exclusion and at the one the configuration asks for. Seven ranks, three seeds, the three observers with the lowest error, median over seeds; the error is against the measured PR^{pos} and is not comparable with table 5, which is taken over more observers. Only the difference between the columns should be read.

r	measured PR^{pos}	MG at $W_T = 150$	MG at $W_T = 624$
2	2.00	2.35	2.24
5	5.00	4.53	4.11
8	8.00	5.82	5.34
11	11.00	9.16	8.67
14	14.00	9.56	9.04
17	17.00	13.16	12.71
20	19.99	13.64	13.45
median absolute error		2.78	3.13
Spearman ρ		1.00	1.00

Three things follow. The ordering, the claim section 5.4 makes at this range, is unaffected: the rank correlation is 1.00 under both exclusions. The absolute value moves down by 0.11 to 0.52 components, so the capped exclusion was mildly inflating it, in the direction that a shortage of exclusion predicts. And the correction makes the recovery slightly worse rather than better, so the cap was not improving the reported result: enforcing the protocol costs accuracy and gains correctness. The saturation above about six directions is present at both settings and is therefore not an artefact of the exclusion.

The same cap has a second effect that the identifiability ratio inherits. The exclusion actually applied is $W_T = \min(\max((E-1)\tau, t_{\text{acf}}), 150)$, and the ratio’s numerator is computed at $2E_{\text{max}}$ with τ held fixed. At the eight-direction configuration the numerator’s span is therefore $(40-1) \times 4 = 156$ and the denominator’s 76: the numerator is capped at 150 whatever the data does, while the denominator is capped only once t_{acf} reaches 150. The two are treated identically exactly when $t_{\text{acf}} \geq 150$ and differently when it is smaller.

Five of the seven parameter-norm records of section 7 are clean by this test. The asymmetric cases are the other two and, more consequentially, the calibration systems of section 5, whose autocorrelation time is one to three samples, so that the reference band of section 6.4 was itself measured with 76 against 150. The effect is small at this span and the diagnostic separates the regimes of section 6 without overlap, but a ratio built from two differently excluded estimates is not the clean quantity section 3.4 describes, and lifting the cap is the first thing to change in any follow-up.

O RUN INVENTORY

Table 24 lists every training run the paper draws on. Two architectures appear: **T** is the one-layer transformer of Nanda et al. (2023), $d_{\text{model}} = 128$, four heads, no layer normalisation, 226 816 parameters on modular addition at $p = 113$ and 228 608 on S_5 , trained in double precision with AdamW at learning rate 10^{-3} and $\beta = (0.9, 0.98)$; **P** is the two-layer quadratic perceptron of Gromov (2023), 145 500 parameters at $p = 97$ and width 500, trained by full-batch gradient descent with no regularisation.

Three departures from the source configurations should be stated, since two of them place the runs in the regimes section 7.1 assigns. Nanda et al. (2023) train full batch; we use mini-batches of 256, which makes this setting stochastic rather than deterministic. The S_5 weight decay of 0.2 and the S_5 task itself (Power et al., 2022) are outside that configuration. And the two S_5 runs step the optimiser

twice per batch, reproducing a duplicated update in our own earlier released logs, so their effective learning rate and decay are about double the nominal ones.

Definition of memorisation and generalisation. Memorisation and generalisation are the first logged step at which the training and the validation accuracy respectively reach 95 %, with no persistence requirement. Training accuracy is measured on the current mini-batch and validation accuracy on a fixed 512-example subset of the validation split, so both carry a sampling error of a few tenths of a per cent, and the step is resolved only to the logging stride, which is 10 steps for the modular runs and 5 for S_5 . The extended reruns of section 7.1 instead require the threshold to hold over a sustained block, which is the stricter rule; the two rules put memorisation within 70 steps of each other on every run here and differ by 1 080 steps on one generalisation step, and no claim turns on either.

A dash means the criterion was not met inside the budget. Nothing here establishes that such a run would never generalise, and table 24 records the final validation accuracy so that the reader can see how far from the threshold each one ended.

Table 24: Training runs used in this paper. α is the training fraction, “final” the validation accuracy at the end of the budget. A dash in the generalisation column marks a run that had not generalised when its budget ended; such runs are censored observations, not established negatives. The name s_5_wd1 denotes the weight-decay member of the S_5 pair, whose decay is 0.2; it is not a decay of 1.

run	task	arch	wd	α	budget	mem.	gen.	final	used in
<i>Trajectory sketch attached</i>									
mod_wd1	mod. add. $p=113$	T	1.0	0.30	20k	1 470	13 700	1.00	section 7.2
mod_wd1_s43	mod. add. $p=113$	T	1.0	0.30	20k	1 580	6 660	1.00	section 7.2
mod_wd1_s44	mod. add. $p=113$	T	1.0	0.30	20k	1 410	3 680	1.00	section 7.2
s_5_wd1	S_5 composition	T	0.2	0.50	15k	1 280	6 735	1.00	section 7.2
mod_wd0	mod. add. $p=113$	T	0.0	0.30	20k	630	—	0.03	section 7.2
s_5_wd0	S_5 composition	T	0.0	0.50	15k	1 015	—	0.02	section 7.2
<i>Extended reruns, 120 000 steps</i>									
gropkpos_s0	mod. add. $p=113$	T	1.0	0.30	120k	1 450	5 110	1.00	section 7.1
lowdata15_s0	mod. add. $p=113$	T	1.0	0.15	120k	620	110 940	0.99	section 7.1
lowdata15_s1	mod. add. $p=113$	T	1.0	0.15	120k	630	—	0.94	section 7.1
lowdata15_s2	mod. add. $p=113$	T	1.0	0.15	120k	660	—	0.01	section 7.1
lowdata20_s0	mod. add. $p=113$	T	1.0	0.20	120k	800	39 600	1.00	section 7.1
wd0_s0	mod. add. $p=113$	T	0.0	0.30	120k	610	—	0.09	section 7.1
wd0_s1	mod. add. $p=113$	T	0.0	0.30	120k	630	—	0.07	section 7.1
p211_wd0	mod. add. $p=211$	T	0.0	0.16	200k	1 800	—	0.04	section 7.1
<i>Perceptron, $p = 97$, full batch, no regularisation</i>									
a_add	$n + m$	P	0	0.5	100k	9 170	11 950	1.00	section 7.1
a_mul	$n \cdot m$	P	0	0.5	100k	9 130	11 600	1.00	section 7.1
a_sub	$n - m$	P	0	0.5	100k	9 040	11 570	1.00	appendix M
a_sq_sum	$n^2 + m^2$	P	0	0.5	100k	8 360	10 380	1.00	appendix M
a_sum_sq	$(n + m)^2$	P	0	0.5	100k	7 280	8 150	1.00	appendix M
x_mix_quad	mixed quadratic	P	0	0.5	100k	9 630	—	0.01	figure 6
x_no_grok	$n^3 + nm^2 + m$	P	0	0.5	100k	9 760	—	0.01	section 7.1
g_p1	$(4n_1 + n_2^2)^3$	P	0	0.5	100k	7 410	12 560	1.00	section 7.1
g_p1x	$+ n_1 n_2$	P	0	0.5	100k	10 050	—	0.01	section 7.1
g_p2	$(2n_1 + 3n_2)^4$	P	0	0.5	100k	6 520	7 750	1.00	section 7.1
g_p2x	$- n_1^2$	P	0	0.5	100k	9 780	—	0.02	section 7.1
g_p3	$(5n_1^3 + 2n_2^4)^2$	P	0	0.5	100k	5 880	6 430	1.00	section 7.1
g_p3x	$- n_2$	P	0	0.5	100k	9 430	—	0.73	section 7.1

The top four perceptron rows are repeated with the trajectory sketch attached, for the measurement of appendix J: four runs at full batch and four with mini-batches of 512, and two of the four again at $p = 23$. The eight at $p = 97$ agree with the rows above on their memorisation and generalisation steps to within one or two logged steps, so the comparison of appendix J is one of noise rather than of learning.

Three notes on how the censored rows should be read. `lowdata15_s1` ends its budget at 94 % validation accuracy and still rising, so it is the clearest censored observation in the set rather than a negative. The generalisation step of `lowdata15_s0` is 110 940 under the sustained-block criterion used for the extended reruns and 109 860 under the plain first-crossing rule used elsewhere; nothing in the paper depends on the difference. And `g_p3x` reaches 73 % validation accuracy against a majority-class baseline of 1.3 % while being provably outside the class the architecture can represent, so it is the one row in this table that neither of the source papers accounts for.

P THE THEILER EXCLUSION AND THE DIMENSION OF A TRANSIENT

Section 3.3 claims that recurrence is a requirement of this protocol and not of the estimator. The claim is testable on the excitation cases of section 5.4 by moving one setting and nothing else: the Theiler exclusion W_T , from zero to the frozen rule. Four observers, one per family, three seeds, $r \in \{2, 4, 6, 8\}$, seven windows each; the estimator is otherwise the frozen configuration, and at the frozen setting every cell reproduces section 5.4 to every stored digit. Table 25 gives the result.

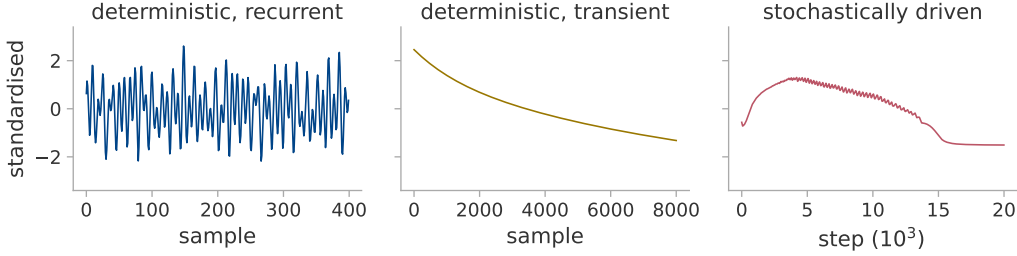


Figure 10: What the estimator actually reads in each regime. The three regimes of table 1 as raw scalar logs, each standardised as the estimator standardises it, since that is all the estimator is given. Left, the recurrent case of this appendix: a quasiperiodic drive of period sixteen samples over twenty-five of them, whose orbit closes onto a torus. Middle, the transient case over its whole record — a decay that increases on none of its eight thousand steps. Right, a real mini-batch transformer run through the parameter norm, which is the stochastic regime. The last two are the ones the diagnostics of section 3.4 must refuse, and the difference between them is why one statistic will not do it: the transient is monotone and the driven run is not, so a test of non-monotonicity catches the first and a test of embedding stability the second.

Table 25: The estimate against the Theiler exclusion, median over observers, seeds and windows. The transient’s estimand is 1: at $W_T = 0$ the estimator returns it, and the exclusion is what removes it. The recurrent cases are unchanged under the same variation, and 28.96 is not a level but the value a divergent quantity happens to take at the implementation’s cap of 150.

W_T	0	2	5	20	100	150	uncapped
recurrent, fast	5.39	5.42	5.42	5.42	5.42	5.41	5.41
recurrent, slow	2.11	2.92	3.23	3.32	3.33	3.33	3.33
transient	1.20	1.71	2.38	5.51	20.83	28.96	39–172

Three things follow. *The number at $W_T = 0$ is the right one.* On a uniformly sampled curve the 20 nearest neighbours lie at offsets $\pm 1 \cdots \pm 10$, for which equation (3) returns 1.19928 exactly; the transient case returns 1.1971 ± 0.0010 across all forty-eight cells. The excess over unity is the lattice bias of a deterministically sampled curve, not an error, and the measured trajectory participation ratio of the same case is 1.03 to 1.07.

The mechanism is the loss of near neighbours, not of neighbours. The estimator fills its twenty slots at every exclusion; what changes is how far away they are. Before any exclusion the median temporal separation of a point’s twenty nearest neighbours is 2546 samples on the fast case and 6 on the transient — there, the nearest neighbours in space simply *are* the temporal ones. Excluding

150 samples costs the fast case almost nothing, 138 048 close pairs surviving of 138 761, and costs the transient all 131 849 of its own. Its twenty retained distances then span a factor of 1.07, their log-ratio sum falls from 15.86 to 0.63, and the reciprocal in equation (3) diverges. No cell is marked degenerate, so this is not a numerical floor. An exponential decay with no optimiser and no data reproduces both ends, 1.196 and 26.4.

Two qualifications. Recurrence is not always needed even here: in the fast case $W_T = 0$ already returns r , because it is not oversampled, and it is the slow case — nearest neighbours 574 samples away — that the exclusion rescues, from a rank-blind 2.1 to its published 3.5. And a transient need not have dimension one: a decaying *oscillation* returns 2.36 at every exclusion, correctly, because at $k = 20$ the neighbourhood spans several turns of the spiral and the two-dimensional sheet they lie on is all there is to see at that resolution. The asymmetry is general; the value is not.

A recurrence-based diagnostic. $\hat{d}_{MG}(W_T=150)/\hat{d}_{MG}(W_T=20)$ is 1.000 on the fast case and 1.001 on the slow, against 5.25 on the transient, with no overlap over forty-eight cells per case (largest recurrent 1.012, smallest transient 4.71); the less expensive $\hat{d}_{MG}(100)/\hat{d}_{MG}(50)$ separates as cleanly. Unlike the trend-crossing count it tests recurrence in the reconstructed space rather than non-monotonicity of the series, which is the gap section 3.4 leaves open and which appendix Q shows costs something on real runs. We report it rather than adopt it: it is calibrated on the same systems that produced the two diagnostics and has not been validated out of sample. The principled alternative is older than either. A space–time separation plot (Provenzale et al., 1992) shows directly how neighbour distance depends on temporal separation, and reading W_T off one would replace the rule of section 3.2 with a measurement; the variation above is a coarse version of that plot, and choosing the exclusion this way rather than by rule is the obvious improvement to the protocol.

Q DESCENT AT THE EDGE OF STABILITY

Every system of section 5 is driven, so the ladder establishes a conditional, and section 6.1 asks whether anything undriven occupies the admissible regime. The one candidate the literature offers is the edge of stability (Cohen et al., 2021): gradient descent at a fixed rate sharpens until the top Hessian eigenvalue reaches $2/\eta$ and then hovers, and while it hovers the loss descends on long scales and not on short ones. That is deterministic and not a plain transient, which is the pair of conditions section 3.3 asks for.

Setting. Full-batch descent, no weight decay, on the quadratic perceptron of appendix F: modular addition at $p = 97$, width 500, half the pairs held out, eight rates by two seeds, 30 000 steps, summarised in table 26. Two departures are deliberate. The log is written at *every* optimiser step, because the oscillation is the two-cycle of the unstable mode and its period is a few steps, while every other full-batch log here is strided at ten or more. And the sharpness is measured along the run by power iteration on Hessian-vector products, taking the largest *algebraic* eigenvalue: the Hessian is indefinite early in training, when the most negative direction is the steeper, and plain power iteration then returns a negative sharpness. The routine agrees with an exact eigendecomposition on a 120-parameter model to 6×10^{-5} relative at the setting used.

Table 26: Whether undriven descent at the edge of stability oscillates, and whether it recurs. Full-batch descent by learning rate, both seeds. The stability ratio is $\eta\lambda_{\max}/2$ over the second half of the run; *rises* is the fraction of post-transition steps on which the loss increases; *return* is the nearest neighbour surviving a 150-sample exclusion divided by the distance the orbit travels during that exclusion, for which a two-frequency torus scores 0.008 and a monotone decay 1.007. In the middle band the sharpness self-limits at $2/\eta$: oscillation arrives at every rate there, but recurrence only in two runs of seven.

η	stability ratio	rises	return	outcome
1×10^5	0.238, 0.228	0.000	1.007, 1.006	monotone
3×10^5	0.749, 0.707	0.000	1.007, 1.007	monotone
1×10^6	0.978, 0.956	0.450, 0.500	0.065, 1.008	at the edge
1.5×10^6	0.966, 0.965	0.500, 0.500	1.013, 1.013	at the edge
2×10^6	0.976, 0.976	0.500, 0.500	1.013, 1.013	at the edge
2.5×10^6	0.989, —	0.500, —	0.026, —	one seed diverges
2.8×10^6	—	—	—	diverges
3×10^6	—	—	—	diverges

Oscillation without recurrence. The oscillation is real and not round-off: the largest single-step rise is 0.99 to 4.1 times the loss itself, against a float32 relative epsilon of 1.2×10^{-7} . But oscillation is not recurrence. Two of the seven edge-of-stability runs return near states they have left; five trace a curve past the exclusion exactly as the monotone runs do, and the return column of table 26 separates the two groups by an order of magnitude with nothing between them.

Performance of the diagnostics. On the training loss, the observer this paper’s full-batch analysis uses, the admissible rectangle of section 3.4 admits all five runs that do not recur and refuses both that do. The estimate agrees with the recurrence measurement and not with the rectangle: 3.49 and 2.44 in the two runs that return, against 29.2 to 30.8 — the transient value of appendix P — in the five that do not.

The cause is visible in the inputs. Every run at the edge has a rise fraction of 0.50, so the trend-crossing count, which tests non-monotonicity and not recurrence, passes all of them, while ρ_{ident} happens to equal unity in the five and at 1.23 and 1.30 in the two. On the parameter norm the rectangle admits all seven. This is seven runs of one configuration on one task and we draw no rule from it, but it is the failure mode section 3.4 names, occurring.

Dependence on the logging stride. Read at the stride every other full-batch log here uses, the phenomenon is largely erased: over these seven runs the median trend-crossing count falls from 80 to 2 and the median rise fraction from 0.500 to 0.000 exactly, and where five of the seven were admissible at every step, none is at a stride of ten and one is at fifty. The collapse is not uniform — the run with the most intermittent oscillation keeps a rise fraction near 0.45 under decimation, because its cycle is not a clean two-cycle to alias — but whether a log looks like a transient is in part a property of how often it was written.

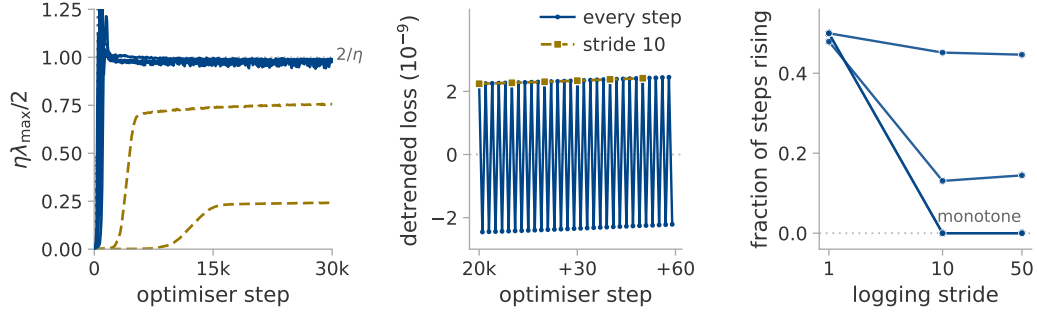


Figure 11: Whether full-batch descent reaches the edge of stability, and what the logging stride removes. (a) The stability ratio along training, seed 1 of each rate; the seeds agree to within the linewidth. Rates that stay below the dotted line descend monotonically and are drawn as transients; those that pin against it are at the edge. (b) Sixty consecutive steps at $\eta = 10^6$, deep in the edge-of-stability phase, against the same series read at a stride of ten: the two-cycle is not blurred by that stride but removed by it. (c) The fraction of steps on which the loss rises, for each edge-of-stability rate, against the stride the log is read at.

R THE CEILING: EMBEDDING DIMENSION AGAINST RECORD LENGTH

Section 5.2 reports that the estimator saturates. Two explanations were available and the paper could not choose between them: the embedding condition $E > 2d$, which at $E_{\max} = 20$ allows ten components, and the finite-record bound $D \lesssim 2 \log_{10} N$ (Eckmann & Ruelle, 1992), which at $N = 8000$ allows 7.8. They are separable because each is insensitive to the other’s knob. We vary $E_{\max} \in \{10, 14, 20, 28, 40, 56\}$ at $N = 8000$, and $N \in \{10^3 \dots 6.4 \times 10^4\}$ at $E_{\max} = 20$, over $r \in \{2 \dots 20\}$ and three seeds with everything else frozen; table 27 gives both. Each N is a *prefix* of one longer record, not a resampling of a fixed span, so the delay lag keeps its meaning in periods; all twenty-seven trajectories are bit-identical to the published ones over their common prefix, and rescoring the published twenty-direction configuration returns 15.09 against its published 15.1.

Table 27: Whether the ceiling follows the embedding condition or the finite-record bound. Two scans, three seeds per cell. *Tracking* is the largest r at which the median estimate stays within one component of the truth; *level* is the median estimate at $r = 20$; *slope* is dd_{MG}/dr over $r \geq 8$, which is 1 if the estimator still tracks the rank and 0 if it has stopped. The predictions each hypothesis makes for the tracking column are given beneath it.

$E_{\max}$ at $N = 8000$	10	14	20	28	40	56	
tracking	6.6	7.4	7.9	8.5	8.6	9.0	
$E_{\max}/2$ predicts	5	7	10	14	20	28	
level at $r = 20$	7.3	8.7	10.5	11.9	13.3	14.5	
slope	0.10	0.16	0.27	0.38	0.50	0.60	
N at $E_{\max} = 20$	10^3	$2 \cdot 10^3$	$4 \cdot 10^3$	$8 \cdot 10^3$	$1.6 \cdot 10^4$	$3.2 \cdot 10^4$	$6.4 \cdot 10^4$
tracking	7.5	8.0	7.6	7.9	8.2	9.3	9.8
$2 \log_{10} N$ predicts	6.0	6.6	7.2	7.8	8.4	9.0	9.6
level at $r = 20$	9.4	9.7	10.2	10.5	10.6	10.8	11.1
slope	0.22	0.23	0.24	0.24	0.25	0.25	0.25

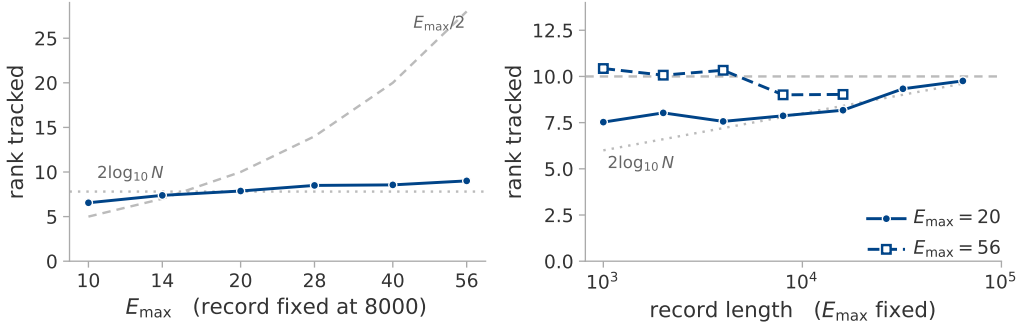


Figure 12: The rank the estimate tracks to, against each hypothesis’s knob and against what that hypothesis predicts. Every series is the constructed recurrent system, so the figure carries one hue and the two predictions are reference lines. (a) Sweeping $E_{\max}$ at a fixed record. The measured ceiling rises from 6.6 to 9.0 while $E_{\max}/2$ rises to 28; the two cross near the $E_{\max} = 20$ this paper froze, which is the coincidence section 5.2 reports. (b) Sweeping the record at fixed $E_{\max}$. The dotted line is $2\log_{10} N$; the dashed line is $E_{\max}/2$ for the filled series only, the open series having a Takens prediction of 28 that is off scale. The open series is the control that decides it: at $E_{\max} = 56$ the estimate still tracks to 10.4 on a record of one thousand samples, where a finite-record bound allows 6.0.

Outcome of the two scans. The level at high rank is the embedding’s: sixty-four times the record adds 1.7 components, where a factor of 5.6 in $E_{\max}$ adds 7.2. The tracking ceiling is neither’s. It never passes 10.4 over the sixteen distinct $(E_{\max}, N)$ cells — the two scans above plus a third that repeats the record scan at $E_{\max} = 56$ — and the cell that attains it is $E_{\max} = 56$ at $N = 1000$, the *shortest* record in the study, where $2\log_{10} N$ allows 6.0. A record-length bound cannot produce that, so Eckmann & Ruelle (1992) is not the explanation here. Nor is $E > 2d$ in the form we used it: the ceiling rises by 0.047 components per unit of $E_{\max}$, not by a half, so $E_{\max}/2$ happens to be right at $E_{\max} = 20$ and is wrong at both ends — which is Sauer & Yorke’s point in the other direction, that how many delay coordinates one needs does not follow from the embedding theorem alone (Sauer & Yorke, 1993).

Fitting the level at $r \geq 16$ separates the candidates outright (table 28). Neither hypothesis fits; their pointwise minimum fits no better, so they do not jointly bind either; and a form logarithmic in both knobs fits an order of magnitude inside the data’s own spread. Both dependences are therefore logarithmic: 2.6 components per doubling of $E_{\max}$, 1.0 per decade of record.

Table 28: Whether either candidate explanation of the ceiling accounts for the measured level. Candidate forms fitted at $r \geq 16$ over the sixteen $(E_{\max}, N)$ cells; the spread of the fitted quantity itself is 2.01 components, so a form with a larger error than that describes nothing.

form	root-mean-square error
$E_{\max}/2$, the embedding condition	8.54
$2\log_{10} N$, the finite-record bound	3.91
the pointwise minimum of the two	3.95
$8.63\log_{10} E_{\max} + 1.03\log_{10} N - 5.52$	0.19

Two qualifications. First, $E_{\max}$, τ and the delay span $(E_{\max} - 1)\tau$ are algebraically locked, and a third scan holding the span at 76 while τ falls from 8 to 1 finds that $E_{\max}$ then adds nothing (regression slope 0.000). The operative variable looks like the temporal extent of the delay window, which is not what $E > 2d$ is about either; that scan is defective at both endpoints, so it does not settle the converse.

Second and more seriously, *the linear null reproduces the whole dependence*. PR_{delay} , a spectral statistic of the delay covariance with no geometry in it and capped only by E , rises from 5.33 to 13.53 across the $E_{\max}$ scan where MG rises from 6.94 to 13.55, and is flat to two decimals across

the record scan. It is fitted by $11.09 \log_{10} E_{\max}$ with a root-mean-square error of 0.09, the tightest fit anywhere in this appendix. Most of what we have been calling a ceiling is therefore how many components the delay window resolves *linearly*, which is a spectral reading of the saturation and not a geometric one.

The unexplained tracking limit. Why the tracking ceiling lies near ten components. It is not $E_{\max}/2$, since it is 10.4 at $E_{\max} = 56$; not $2 \log_{10} N$, since it is 10.4 at $N = 1000$; and not the measured effective rank, which equals r to 0.01 at every $N \geq 2000$. No form we fitted accounts for it, and the controls are clean — the Theiler axis is inert (a fixed-76 scan reproduces the frozen setting to two decimals at every cell), and none of the 3240 cells is marked degenerate. This is a real limit of the instrument that we can bound but not name.

Section 3.4’s own ratio does see it: ρ_{ident} at $r = 20$ is 1.31 to 1.40 at every record length, a longer record never making an unidentifiable estimate identifiable, and falls from 1.22 to 1.02 as $E_{\max}$ grows, crossing the band at about the rank where tracking stops.

S COMPUTE COST AND CODE AVAILABILITY

Compute cost. Everything here runs on one consumer GPU and one desktop CPU, in hours rather than days. The largest GPU item is the edge-of-stability campaign of appendix Q: sixteen runs of 30 000 full-batch steps, logged at every step and with the sharpness measured three hundred times per run, in 1770 seconds total on a single T4. The analysis is the more expensive half and is all CPU, a few core-hours per experiment, the ceiling scan of appendix R the largest single item. Nothing in this paper needs a cluster, which is deliberate: the instrument is meant to be inexpensive enough to run on a log one already has.

Cost of the trajectory sketch. The claim of section 7.2 that measuring the trajectory directly is affordable should carry a number. In storage it is exact arithmetic: at each logged step the sketch keeps 1024 coordinates in each of two spaces under two hash families, 4096 floats, against the full parameter vectors of appendix O — a factor of 36 for the perceptron and 55 for the transformer. Over the 151 logged steps of a 1500-step run that is 2.5 against 88 megabytes.

In time the overhead is below what we can measure. Training the same configuration with and without the observer attached, alternating the order so that warm-up cannot favour either, and checking that the two agree on every logged loss to the last bit, gives 41.75 seconds bare against 40.44 probed — a nominal overhead of -3.1% . A negative overhead is not a result but a bound: at the logging stride the transformer runs use, the cost of the sketch is under the run-to-run variation of the machine, which here is a few per cent. `sketch_cost.py` records the measurement.

Code availability. All code, configurations and result files behind every number and figure in this paper will be released publicly on acceptance: the estimator and its frozen configurations, the six constructed systems, the trajectory sketch, the training scripts for both grokking settings, and the analysis scripts named throughout these appendices. The figures regenerate from the committed result files with a single command.